Geometry and Motion: Stereo, Structure from Motion

Lecture 16
Notes for Probabilistic Models of the Visual Cortex

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What this lecture covers

Projection and vanishing points

Calibration and stereo

Structure from motion

Geometric priors

Summary

Unless a caption says otherwise, the figures use colors consistently. In the projection figures the camera axes $\vec{a}, \vec{b}, \vec{c}$ are red, blue and teal. In the stereo figures the first camera is blue and the second red, so scene points are drawn in other colors. Orange marks image points, projections and recovered shapes; purple marks vanishing points and normals; teal marks epipolar lines and planes. In the Manhattan figures the three axes are red, blue and teal, and olive marks random edges. Captions that cite the 2020 slides took their pictures from them (Yuille, 2020); every other figure is computed with the models described in the text.

Most analysis of projection uses the pinhole camera model

An image is formed by light rays that travel from the scene to the camera. In a pinhole camera the light enters through a single point, the pinhole: a pinhole camera has a lens of size zero. Every point of the scene then sends exactly one ray through the pinhole, and the image point is where that ray meets the imaging plane (also called the image, view, projection or picture plane). This is perspective projection. Most analysis of projection is based on this model.

Real cameras have lenses of finite size, which bend the rays. Szeliski (2022) describes the corrections to the pinhole model, which are often required near the boundaries of images.

The 2020 slides illustrate the model twice. The first is the classic diagram: rays from an object pass through the center of projection and cross the view plane. The second is the viewing frustum of computer graphics (OpenGL, the Open Graphics Library): the pyramid of space that is drawn, cut off by a near and a far plane. The eye looks along the $-z$ axis and the image plane is drawn in front of the eye, so the image is upright. The angle fovy (field of view in y) is the vertical field of view.

Most analysis of projection uses the pinhole camera model

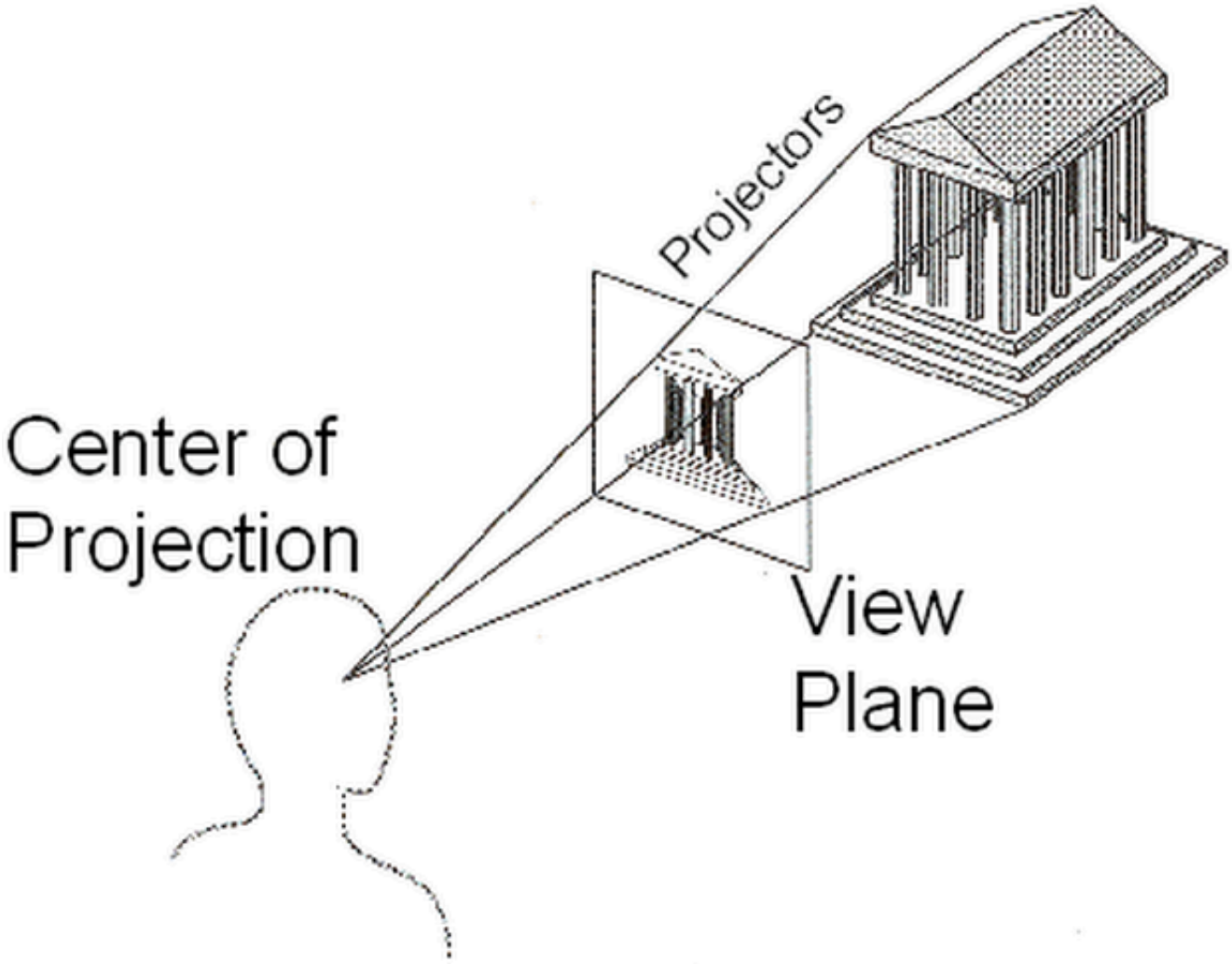


Figure 1: Perspective projection: rays from an object through the center of projection cross the view plane (2020 slides, slide 3).

The viewing frustum of computer graphics

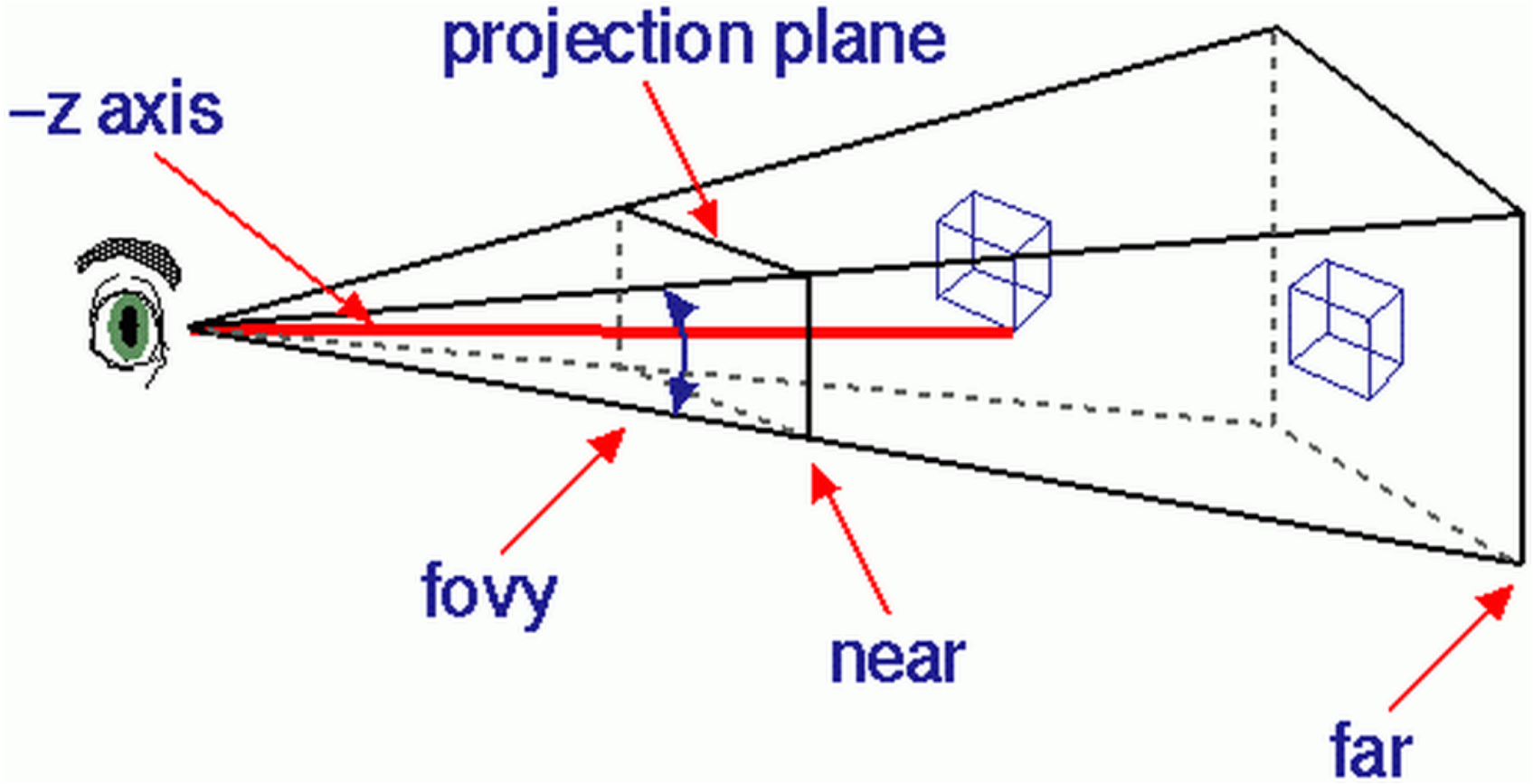


Figure 2: The viewing frustum of OpenGL (2020 slides, slide 3): the eye looks along $-z$; fovy, near and far set the vertical field of view and the range of depths drawn.

Perspective: straight lines stay straight, parallels converge

Perspective projection has two major properties. Straight lines in space project to straight lines in the image. Parallel lines in space project to lines that converge to one point of the image, a vanishing point, which may be at infinity; then the image lines stay parallel. Painters exploited both properties. In Perugino's fresco the lines of the pavement, parallel on the ground, converge at the central door of the temple.



Figure 3: Detail of Pietro Perugino, *Delivery of the Keys to Saint Peter*, fresco, Sistine Chapel, 1481–1482 (2020 slides, slide 4).

A building in two-point perspective

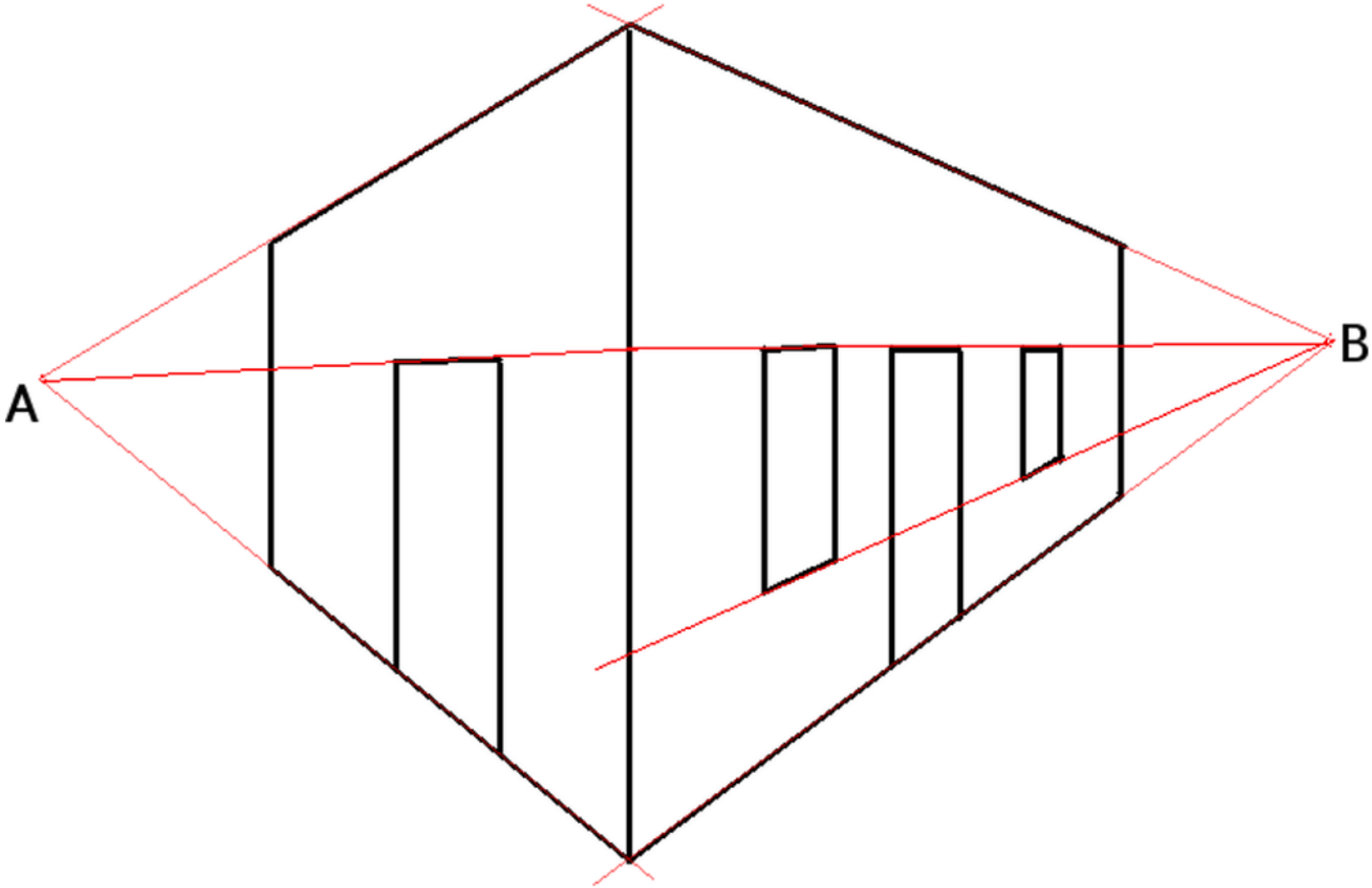


Figure 4: The horizontal edges of a building form two families of parallel lines; each family, extended (red), converges to one vanishing point, A or B. The vertical edges are parallel to the picture plane and stay parallel (2020 slides, slide 4).

A camera is an origin, three axes and a focal length

Describe the camera by an origin $\vec{o}$, at the pinhole, and right-handed coordinate axes $\vec{a}, \vec{b}, \vec{c}$: unit vectors, perpendicular to each other, with $\vec{a} \times \vec{b} = \vec{c}$. The axis $\vec{c}$ is the direction of gaze of the camera; $\vec{a}$ and $\vec{b}$ are the axes of the imaging plane, $\vec{a}$ pointing right and $\vec{b}$ down. The focal length f is the distance from the pinhole to the imaging plane, which lies behind the pinhole:

$$(\vec{r} - \vec{o}) \cdot \vec{c} = -f.$$

A point $\vec{r}$ in space projects to the image point with coordinates (u, v) along $\vec{a}$ and $\vec{b}$ in the imaging plane,

$$\frac{u}{f} = -\frac{(\vec{r} - \vec{o}) \cdot \vec{a}}{(\vec{r} - \vec{o}) \cdot \vec{c}}, \qquad \frac{v}{f} = -\frac{(\vec{r} - \vec{o}) \cdot \vec{b}}{(\vec{r} - \vec{o}) \cdot \vec{c}}.$$

The denominator $(\vec{r} - \vec{o}) \cdot \vec{c}$ is the depth of the point, its distance in front of the camera along the gaze. Because the imaging plane is behind the pinhole, the image is inverted, and that is the origin of the minus signs: a point above and to the right of the gaze lands below and to the left.

A camera is an origin, three axes and a focal length

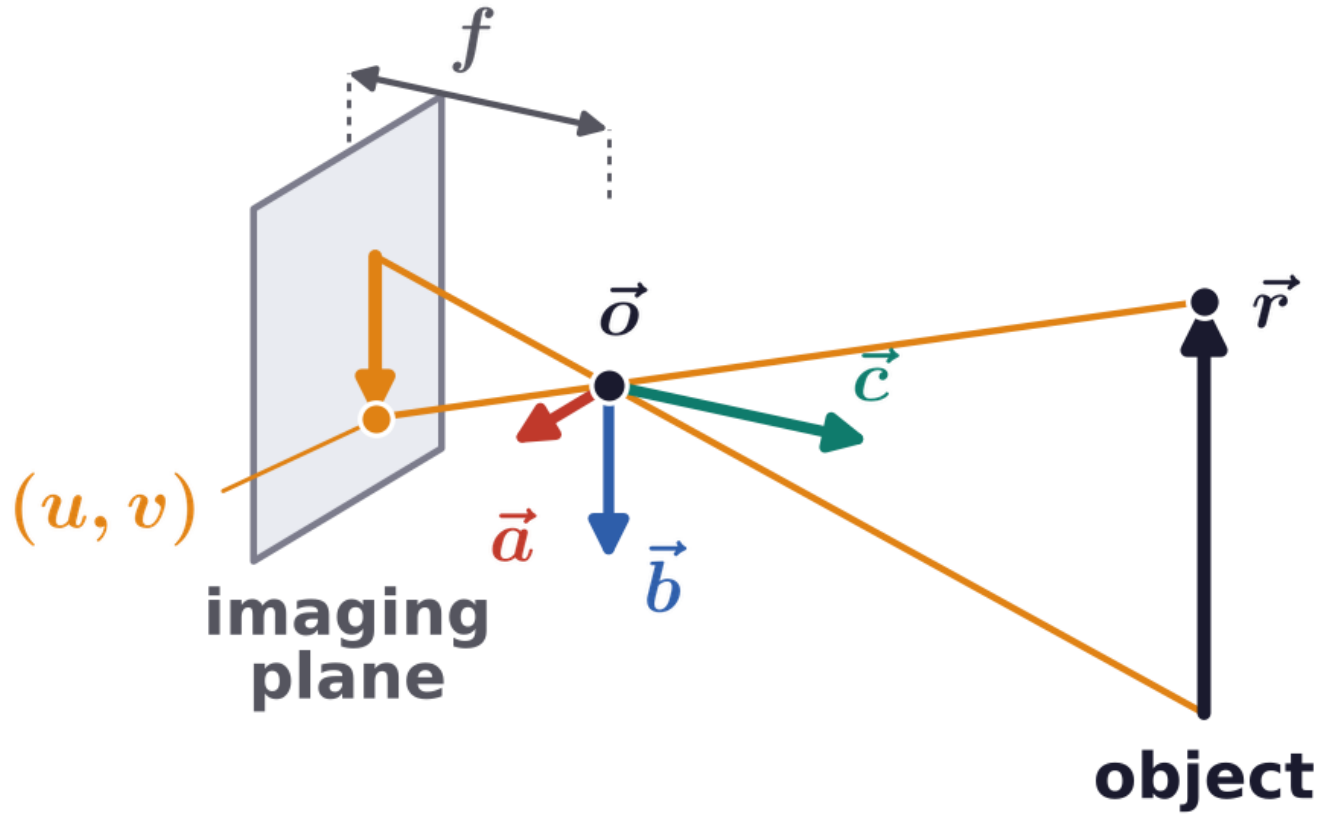


Figure 5: The camera: pinhole $\vec{o}$, axes $\vec{a}$ (red, toward the reader: the camera's right), $\vec{b}$ (blue) and $\vec{c}$ (teal, the gaze), imaging plane at distance f behind. The ray from the object's top (tip $\vec{r}$ of the black arrow) through $\vec{o}$ meets the plane at (u, v): the image (orange) is inverted.

The projection is where the ray meets the imaging plane

For simplicity set $\vec{o} = 0$: the pinhole is the origin, and $(\vec{r} - \vec{o})$ becomes $\vec{r}$. The straight line

$$\vec{r}(\lambda) = \lambda\,\vec{r}$$

passes through the pinhole ($\lambda = 0$) and through the point $\vec{r}$ ($\lambda = 1$). It meets the imaging plane where $\vec{r}(\lambda^*) \cdot \vec{c} = -f$, that is at

$$\lambda^* = -\frac{f}{\vec{r} \cdot \vec{c}}.$$

The point $\lambda^*\vec{r}$ lies in the imaging plane, so it can be written $\lambda^*\vec{r} = u\,\vec{a} + v\,\vec{b} - f\,\vec{c}$. Taking dot products with $\vec{a}$ and $\vec{b}$, which are perpendicular to each other and to $\vec{c}$, gives

$$u = -f\,\frac{\vec{r} \cdot \vec{a}}{\vec{r} \cdot \vec{c}}, \qquad v = -f\,\frac{\vec{r} \cdot \vec{b}}{\vec{r} \cdot \vec{c}}.$$

The division by the depth $\vec{r} \cdot \vec{c}$ is what makes perspective projection nonlinear: the same object looks half as large at twice the distance.

The projection is where the ray meets the imaging plane

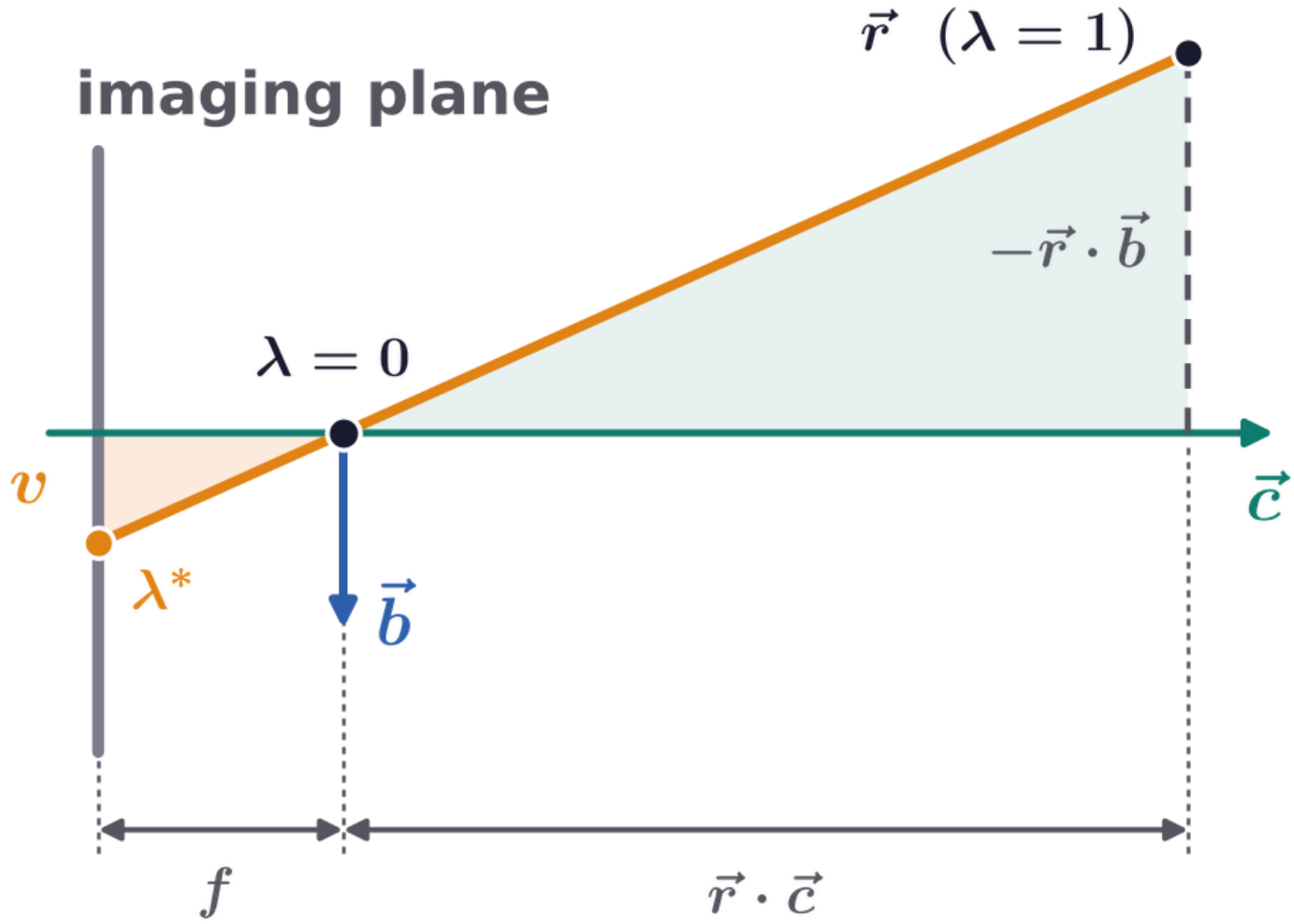


Figure 6: Side view in the plane of $\vec{c}$ (the gaze, to the right) and $\vec{b}$ (down): the ray through $\vec{r}$ ($\lambda = 1$) and the pinhole ($\lambda = 0$) meets the imaging plane at λ^*. The two shaded triangles are similar: v, at distance f behind the pinhole, corresponds to $-\vec{r} \cdot \vec{b}$ at depth $\vec{r} \cdot \vec{c}$ in front.

Straight lines in space project to straight lines

A straight line in space is specified by

$$\vec{r}(\lambda) = \vec{r}_0 + \lambda\,\vec{t}_0,$$

where $\vec{r}_0$ is a point the line passes through (any point on the line can be used), $\vec{t}_0$ is the direction of the line and λ is the position along it. A plane is specified by $\vec{r} \cdot \vec{n} = k$, where $\vec{n}$ is the unit normal of the plane and k is the shortest distance from the plane to the origin.

Why lines stay straight: every ray from a point of the line to the pinhole lies in the plane that contains the pinhole and the line. That plane passes through the origin, so $k = 0$, with normal $\vec{n}$ parallel to $\vec{r}_0 \times \vec{t}_0$. The image of the line is the set of points where these rays meet the imaging plane, which is the intersection of two planes: a straight line. (If the line passes through the pinhole, all its points project to a single image point.)

In the figure the teal triangle is the plane through the pinhole and the line; it cuts the imaging plane in the orange segment, the image of the line. The small panel shows the image upright, as it is usually displayed.

Straight lines in space project to straight lines

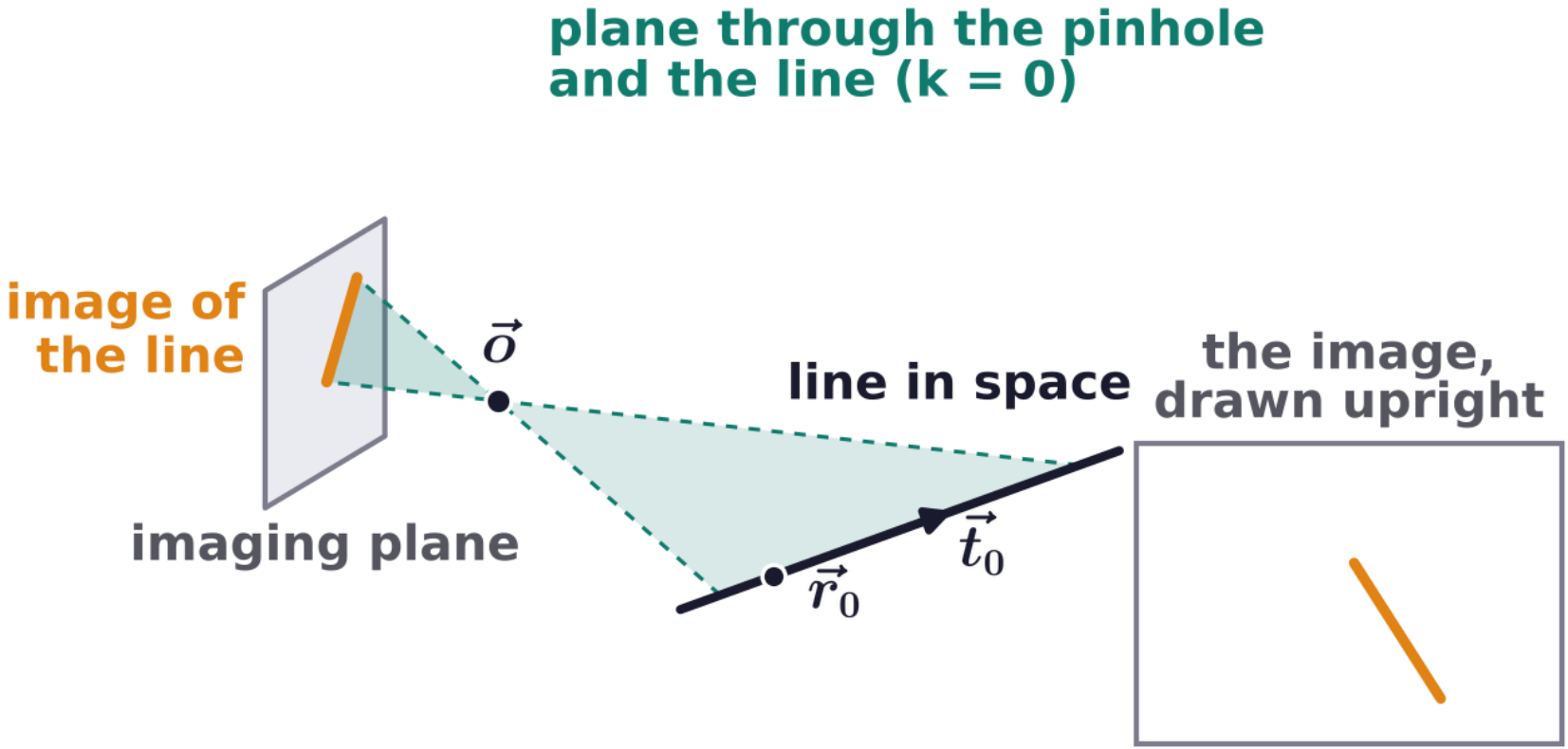


Figure 7: A line in space (black, through $\vec{r}_0$ with direction $\vec{t}_0$), the plane through the pinhole $\vec{o}$ and the line (teal), and the image of the line (orange), which is straight.

The image of a line depends on $1/\lambda$, the inverse distance

Parallel lines in space share one unit direction $\hat{\vec{r}}$; they are written $\vec{r}_0 + \lambda\hat{\vec{r}}$, with λ now the length along the line and one base point $\vec{r}_0$ per line. With $\vec{o} = 0$ each line projects to

$$u(\lambda) = -f\,\frac{(\vec{r}_0 + \lambda\hat{\vec{r}}) \cdot \vec{a}}{(\vec{r}_0 + \lambda\hat{\vec{r}}) \cdot \vec{c}},$$

and $v(\lambda)$ is the same with $\vec{b}$ in place of $\vec{a}$. Slide the base point along its line until $\vec{r}_0 \cdot \vec{c} = 0$, by $\vec{r}_0 \mapsto \vec{r}_0 - \frac{\vec{r}_0 \cdot \vec{c}}{\hat{\vec{r}} \cdot \vec{c}}\hat{\vec{r}}$, which loses no generality; this is possible unless $\hat{\vec{r}} \cdot \vec{c} = 0$, that is unless the lines are perpendicular to the direction of gaze. The denominator is then $\lambda\,\hat{\vec{r}} \cdot \vec{c}$, and

$$u(\lambda) = -f\,\frac{\hat{\vec{r}} \cdot \vec{a}}{\hat{\vec{r}} \cdot \vec{c}} - f\,\frac{\vec{r}_0 \cdot \vec{a}}{\hat{\vec{r}} \cdot \vec{c}}\,\frac{1}{\lambda}.$$

The image position is a constant plus a term in $1/\lambda$, the inverse distance along the line. Equally spaced points therefore crowd together in the image.

The image of a line depends on $1/\lambda$, the inverse distance

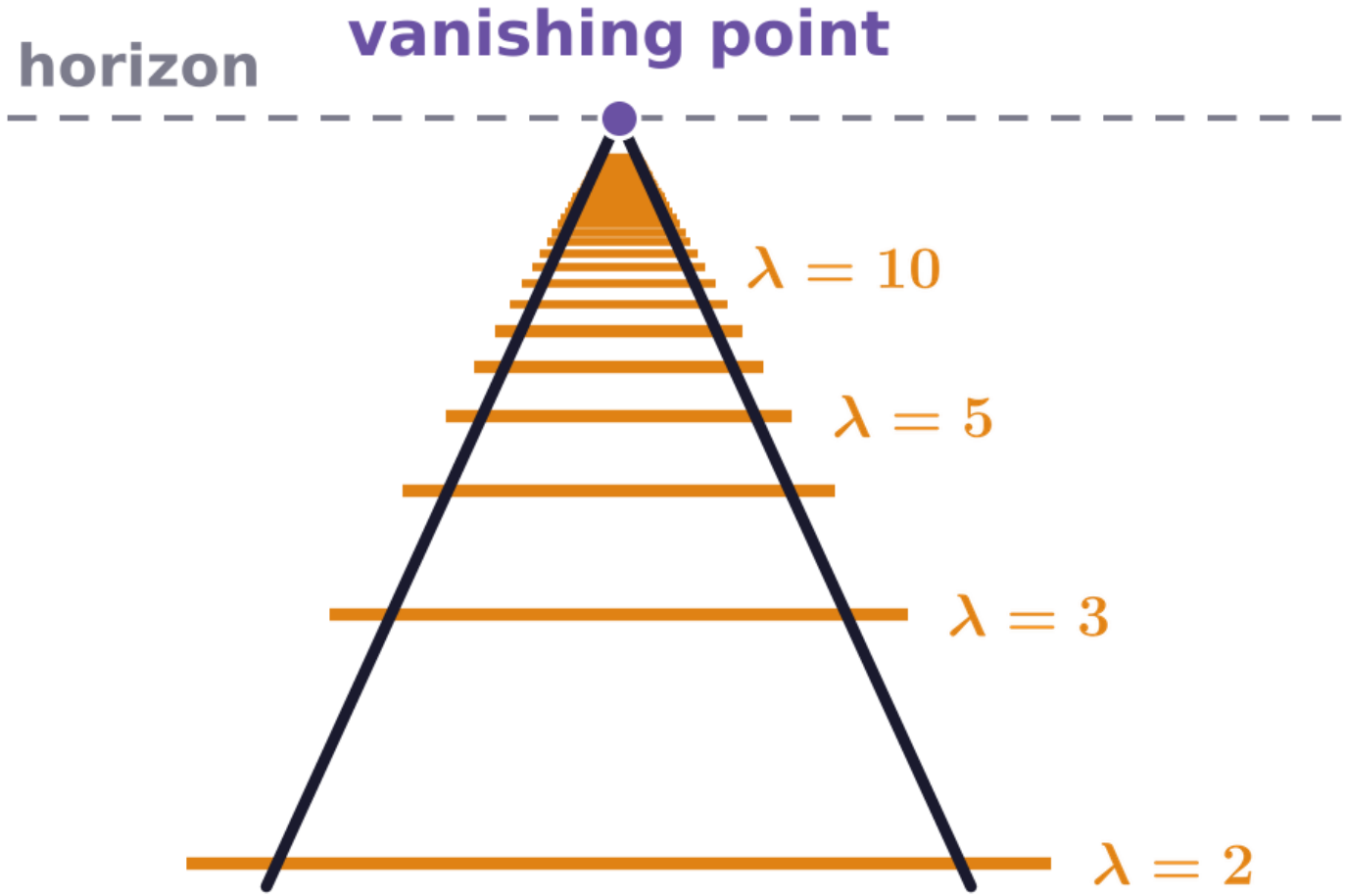


Figure 8: A camera 1.6 m above a railway track, tilted 12° down, sees sleepers (cross ties, orange) every 1 m from $\lambda = 2$ to 40 m (λ from the base point with $\vec{r_0} \cdot \vec{c} = 0$). The positions vary as $1/\lambda$, so the gaps shrink as about $1/\lambda^2$; the rails (black) meet at the vanishing point.

All lines with one direction meet at one vanishing point

Moving along the lines to infinity, $\lambda \to \infty$, the term in $1/\lambda$ vanishes:

$$u \to -f \frac{\hat{\vec{r}} \cdot \vec{a}}{\hat{\vec{r}} \cdot \vec{c}}, \qquad v \to -f \frac{\hat{\vec{r}} \cdot \vec{b}}{\hat{\vec{r}} \cdot \vec{c}}.$$

The limit does not depend on $\vec{r}_0$: all lines with direction $\hat{\vec{r}}$ converge to the same image point, their vanishing point. The exceptions are the lines perpendicular to the gaze, $\hat{\vec{r}} \cdot \vec{c} = 0$: their images are parallel lines, with the vanishing point at infinity.

How many vanishing points an image has depends on how many families of parallel lines the scene contains; some images have none. An image can be given as many as we like: place square tables on a flat floor, none aligned with another. The top of each table has edges in two perpendicular horizontal directions, which give two vanishing points on the horizon, the image line that holds the vanishing points of all horizontal directions. With the camera tilted, as in the figure, all legs are vertical and share one more. So the number of vanishing points is twice the number of tables plus one.

All lines with one direction meet at one vanishing point

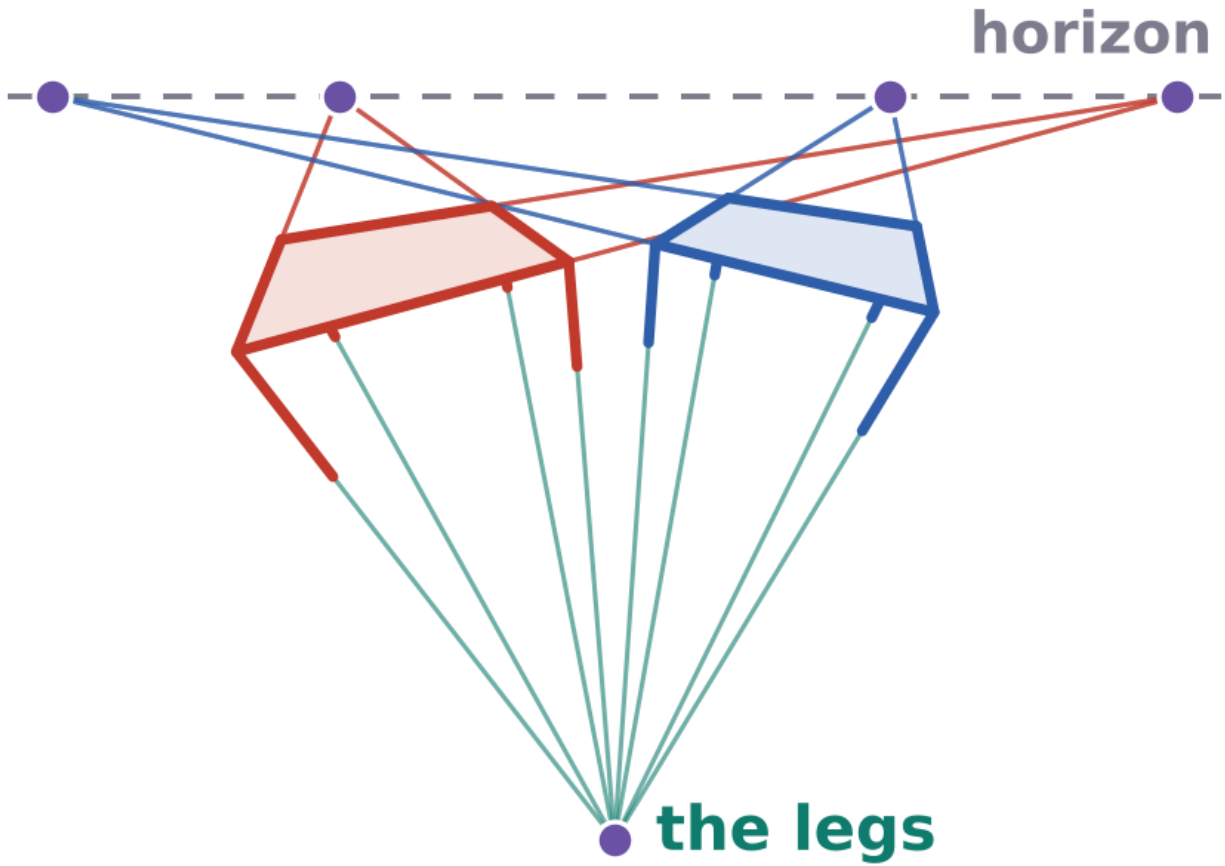


Figure 9: A camera 1.5 m high, tilted 32° down, sees two square tables (red, blue) turned by 35° and 55°: the extended edges of each table meet at two vanishing points (purple) on the horizon, and the extended legs (teal) at one below.

On a ground plane, vertical lines stay parallel in the image

In the real world there is often a flat ground plane, and the direction of gaze $\vec{c}$ is parallel to it. Lines perpendicular to the ground, such as trees growing vertically, are then perpendicular to $\vec{c}$: their vanishing point is at infinity, and their images stay parallel. Ultimately this is physics: humans and animals stand vertically to the ground to resist gravity.

Animals and computer vision systems use the ground plane to estimate the depth of objects. For a camera at height h with level gaze, a point on the ground at depth Z projects $f\,h/Z$ below the horizon: farther feet are nearer the horizon. An object of height H at depth Z has image height $f\,H/Z$, so its size relative to the distance from its feet to the horizon gives its height in units of the camera's height. Filmmakers exploit this assumption to make some actors appear much bigger than others (John Coffey in *The Green Mile*; Gandalf in *The Lord of the Rings*).

In the figure a camera 1.6 m high with level gaze sees three people, 1.8 m tall, at 4, 8 and 16 m, and two trees. In the image their feet approach the horizon as they recede, and the trees stay vertical.

On a ground plane, vertical lines stay parallel in the image

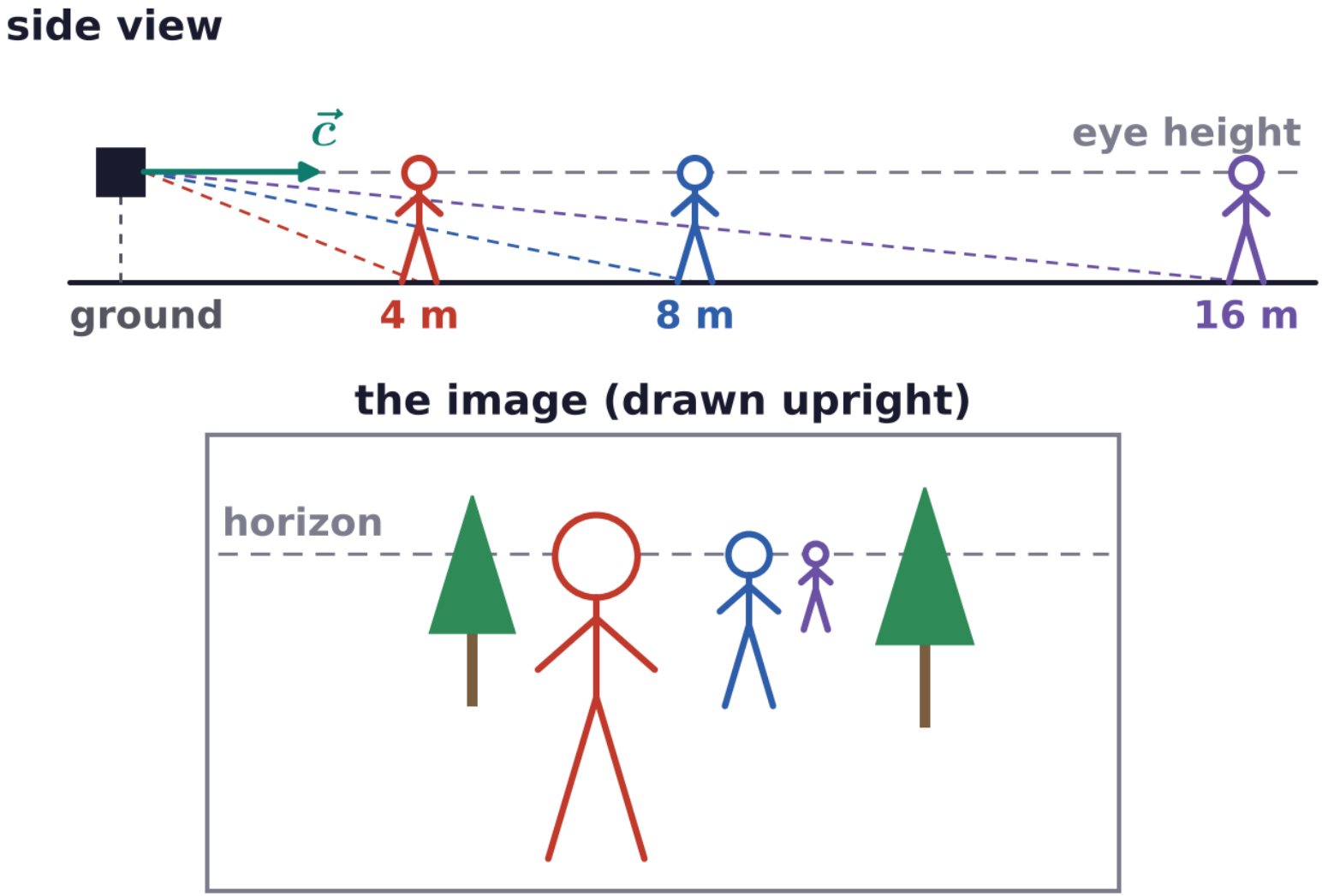


Figure 10: Side view: the camera (black square), its gaze $\vec{c}$ and three people at 4, 8 and 16 m; the dashed lines are the rays to their feet. Below, the image, drawn upright.

Scaled orthographic projection approximates perspective

Perspective projection can often be approximated by scaled orthographic projection, provided the depth $\vec{r} \cdot \vec{c}$ is large and varies little over the viewed scene. Replacing $\vec{r} \cdot \vec{c}$ by a constant Z_0 gives $u = -f\,(\vec{r} \cdot \vec{a})$, $v = -f\,(\vec{r} \cdot \vec{b})$, with f now the scale f/Z_0: orthographic projection (rays parallel to the gaze), then scaling. The projection is linear, hence simpler, but parallel lines stay parallel: the vanishing points are lost. If parallel lines look parallel in an image, a linear approximation is likely valid.

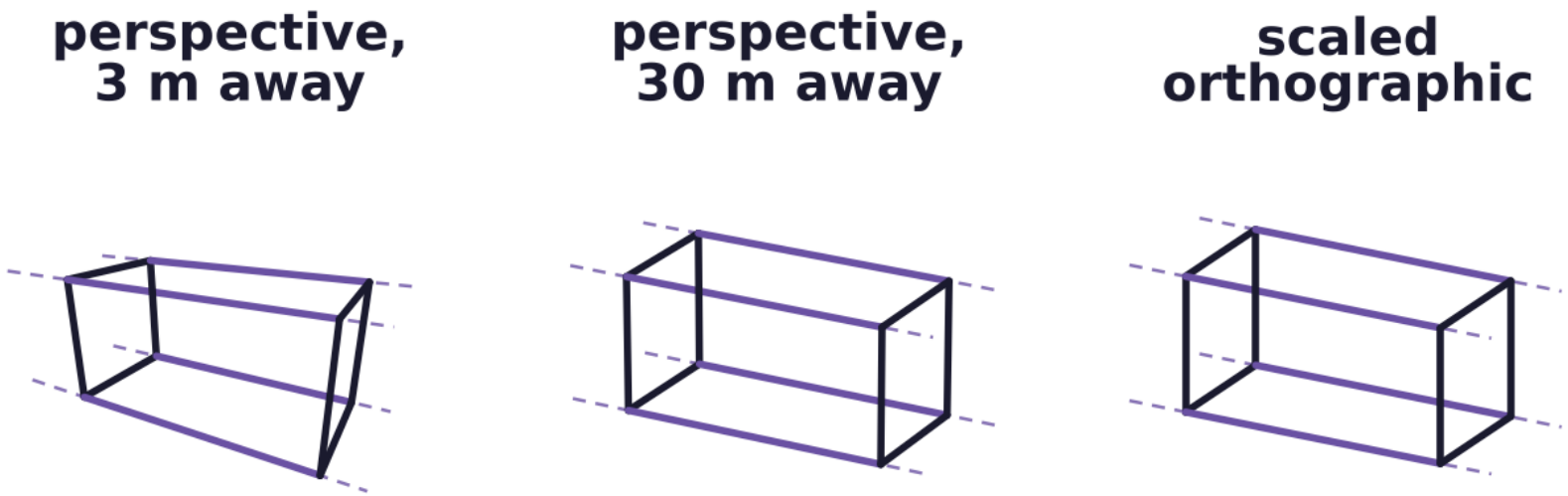


Figure 11: A box from 3 m, from 30 m, and under scaled orthographic projection; purple: four parallel edges, extended (dashed).

Affine cameras are the most general linear projection

The most general linear projection maps a point $\vec{X} = (X, Y, Z)$ to

$$u = K_1 X + K_2 Y + K_3 Z = \vec{K} \cdot \vec{X}, \qquad v = \vec{H} \cdot \vec{X},$$

with two rows $\vec{K} = (K_1, K_2, K_3)$ and $\vec{H} = (H_1, H_2, H_3)$. Scaled orthographic projection, $\vec{K} = -f\vec{a}$ and $\vec{H} = -f\vec{b}$, is the special case $\vec{K} \cdot \vec{H} = 0$ with $|\vec{K}| = |\vec{H}|$: two rows of a scaled rotation. A general affine camera can also stretch and shear the image (Hartley and Zisserman, 2003). People write papers, and entire book chapters, about special cases with restrictions on $\vec{K}$ and $\vec{H}$.

scaled orthographic **general affine**



Figure 12: A cube under a scaled orthographic and under a general affine camera ($\vec{K} \cdot \vec{H} = 0.17$, $|\vec{K}| = 1.04$, $|\vec{H}| = 0.62$).

Binocular stereo estimates depth by triangulation

Two eyes, or two cameras, see each point of the scene along two rays, and the depth of the point is where the rays meet: triangulation. For two parallel cameras a baseline B apart, with the image planes drawn in front of the pinholes (no minus sign), a point at depth Z and lateral position X projects to $u_1 = fX/Z$ in the first image and to $u_2 = f(X - B)/Z$ in the second, so it shifts between the images by the disparity

$$u_1 - u_2 = \frac{f\,B}{Z} :$$

nearer points shift more, and the depth follows from the shift.

Triangulation requires solving the correspondence problem: finding, for a point of one image, the point of the other image that shows the same point of the scene. The epipolar line constraint helps, by reducing the search to a line. Triangulation also needs camera calibration: the relative position and orientation of the two cameras.

Binocular stereo estimates depth by triangulation

In the figure two parallel cameras 0.3 m apart (blue, red) see three points at 2, 3 and 5 m. In the two images, drawn one above the other, each point is displaced by its disparity fB/Z: 0.15, 0.10 and 0.06 focal lengths. The dashed lines join the two images of each point.

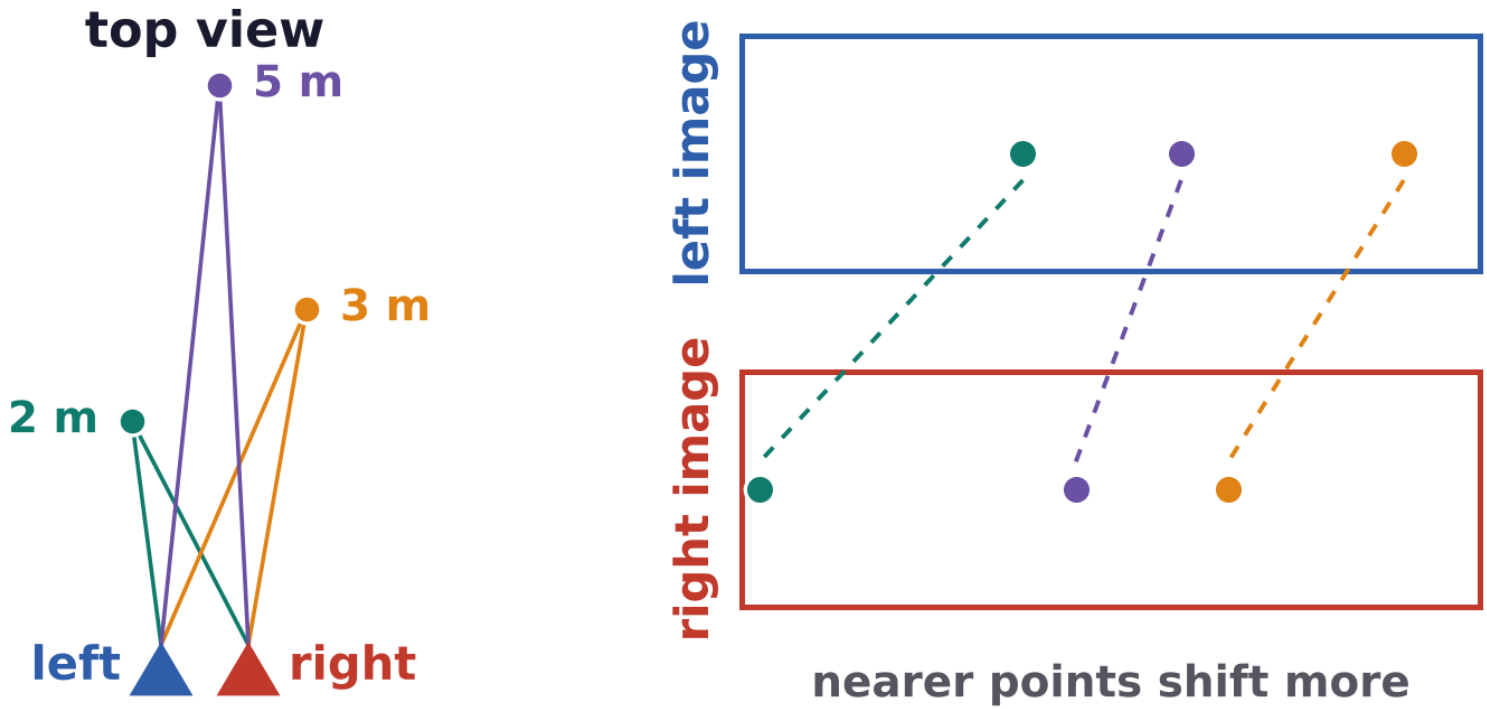


Figure 13: Top view of the two cameras and the three points, and the left and right images: nearer points shift more.

Two eyes fixating a point

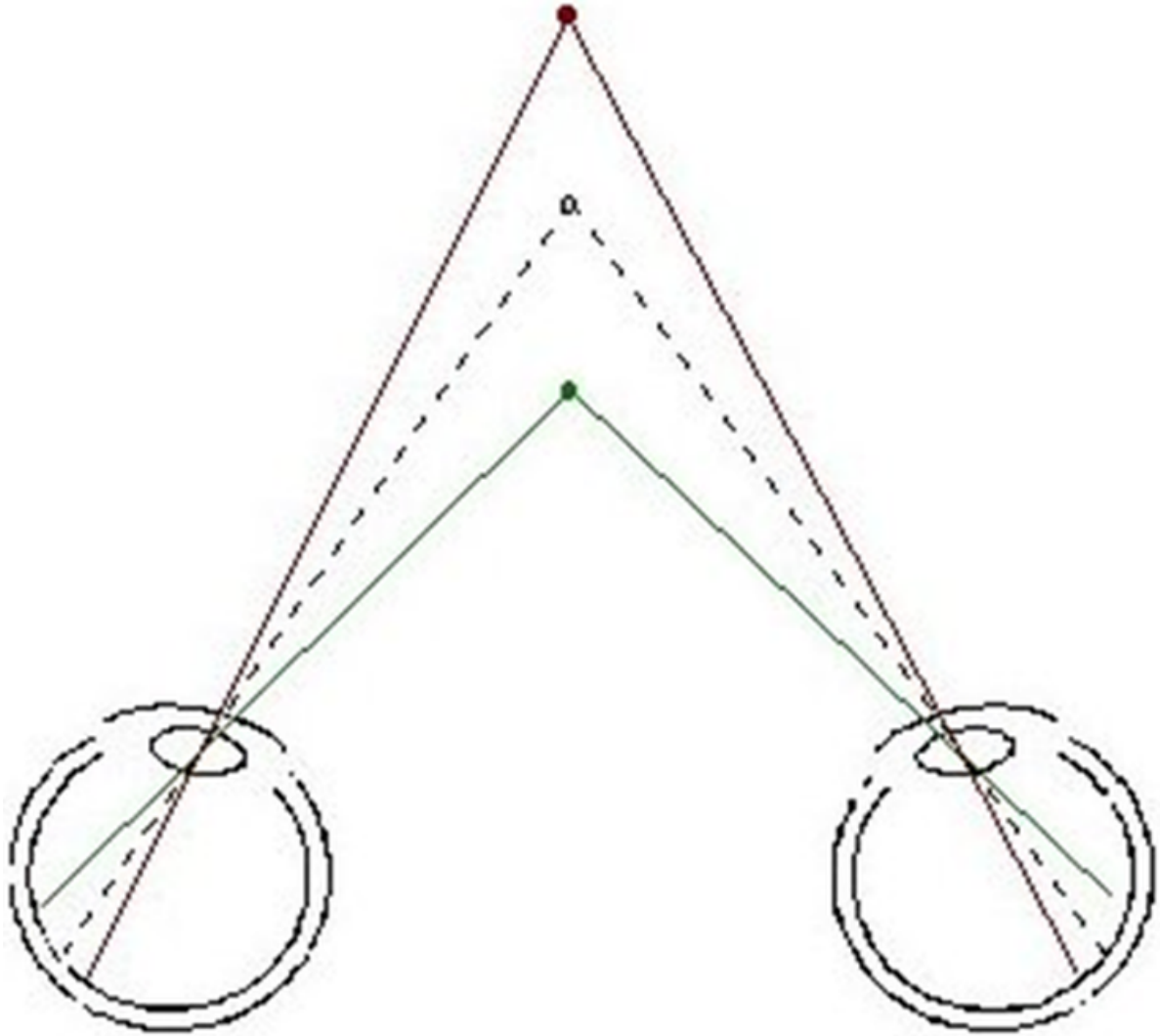


Figure 14: The eyes fixate the point where the dashed lines of sight meet (α: the angle between them). A farther (dark red) and a nearer (green) point land at different places on the two retinas: their disparity (2020 slides, slide 6).

Corresponding points lie on corresponding epipolar lines

A scene point X and the two camera centers (pinholes) C and C' span a plane, the epipolar plane. It cuts each image in a line, an epipolar line. The line CC' joining the camera centers, the baseline, meets the two images at the epipoles e and e'; every epipolar plane contains the baseline, so every epipolar line passes through the epipole of its image.

The image x of X in the first camera only fixes the ray from C through x. All points on that ray lie in the same epipolar plane, so their images in the second camera lie on one epipolar line. Points on one epipolar line can therefore only be matched to points on the corresponding epipolar line, and the search for a match is one-dimensional (Hartley and Zisserman, 2003).

Corresponding points lie on corresponding epipolar lines

In the figure X_1, X_2, X_3 lie on the ray from C through x. The dashed lines from them to C' cross the second image at three points (orange), all on one epipolar line (teal). Purple: the epipoles.

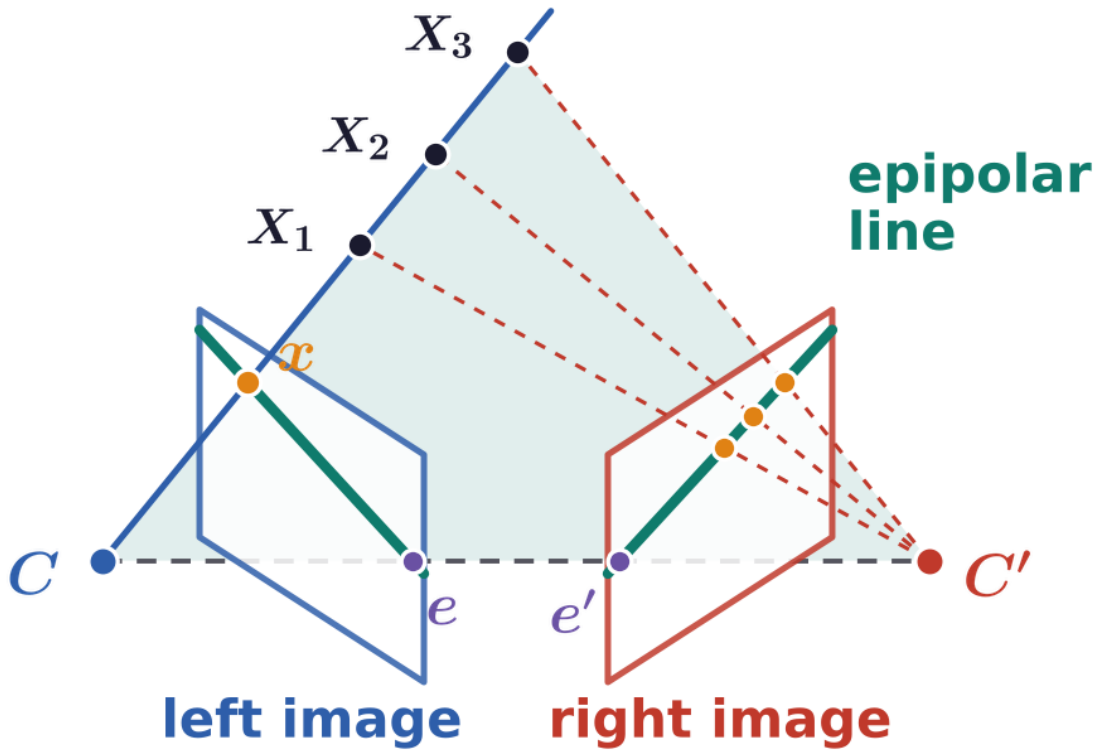


Figure 15: The epipolar plane (shaded) through C, C' and the ray Cx, and the epipolar lines (teal) in the two images.

The epipolar plane and the epipolar lines

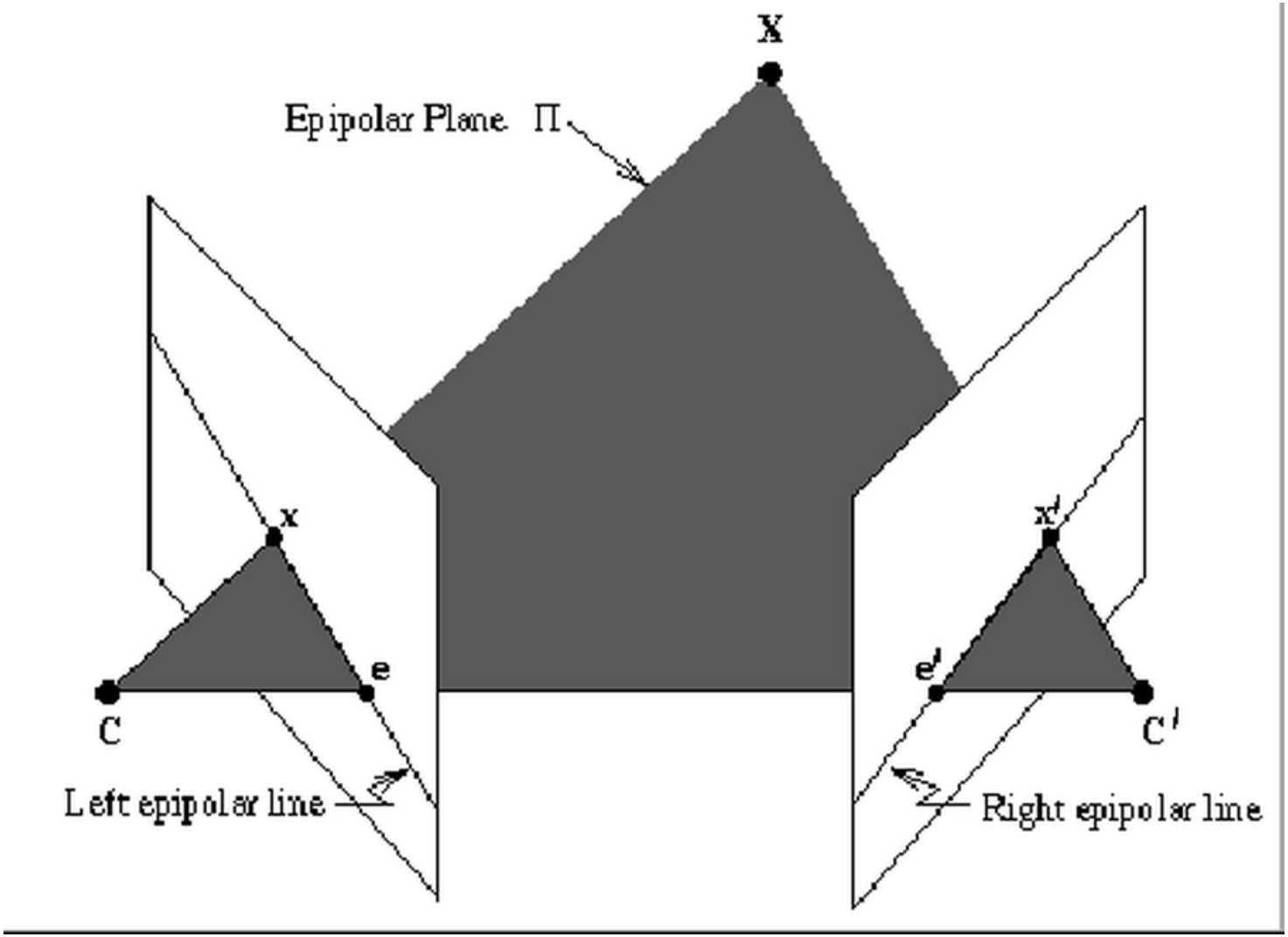


Figure 16: The epipolar geometry of two cameras: the epipolar plane Π through X, C and C' cuts the images in the left and right epipolar lines (2020 slides, slide 7).

Epipolar lines depend on the camera parameters

The epipolar lines depend on the camera positions, orientations and focal lengths. Parallel cameras, side by side with the same orientation, have epipoles at infinity and horizontal epipolar lines: a match lies on the same row. Converging cameras give slanted epipolar lines through the epipole.

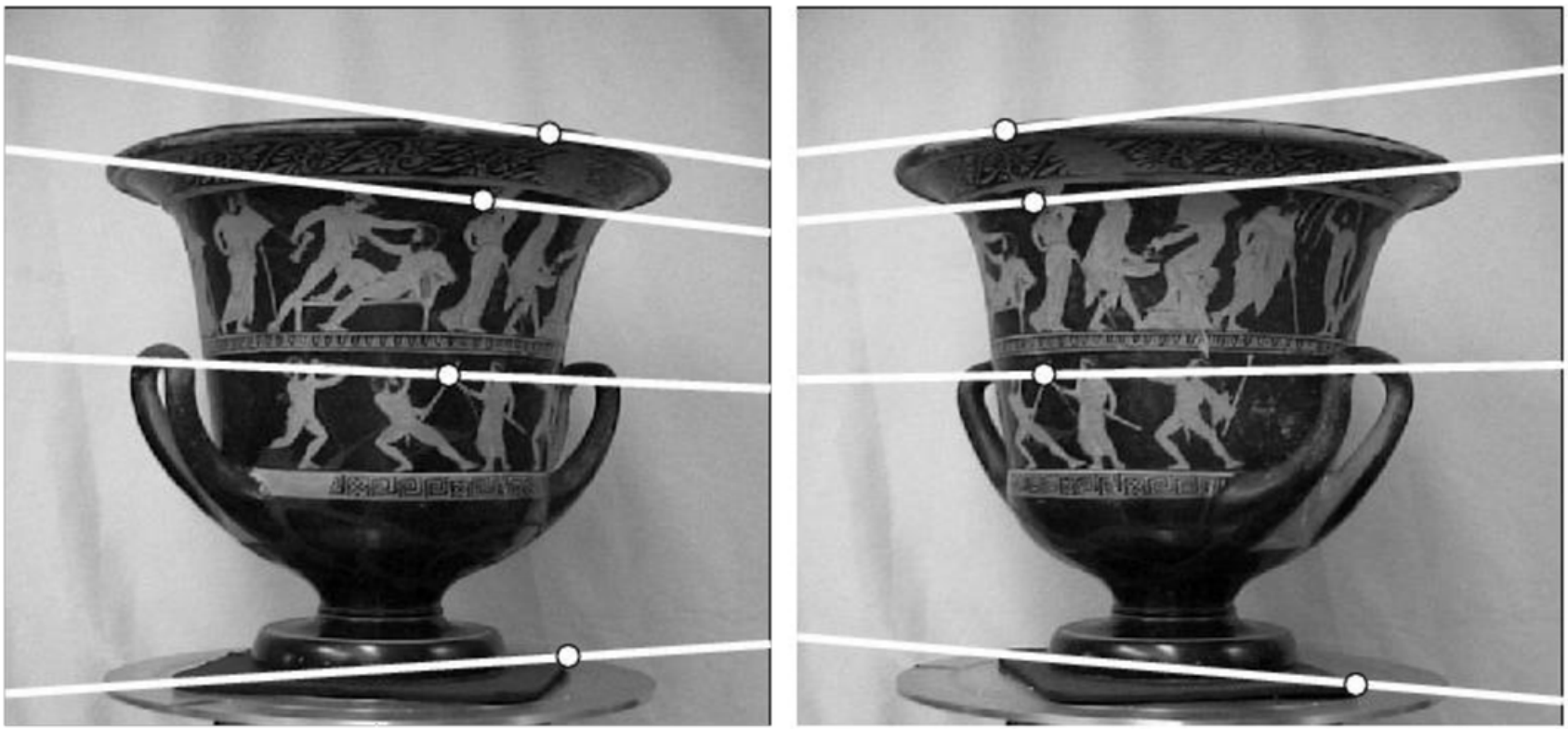


Figure 17: Two views of a vase with corresponding epipolar lines (white) through matched points (circles), from converging cameras (2020 slides, slide 7).

Parallel and converging cameras

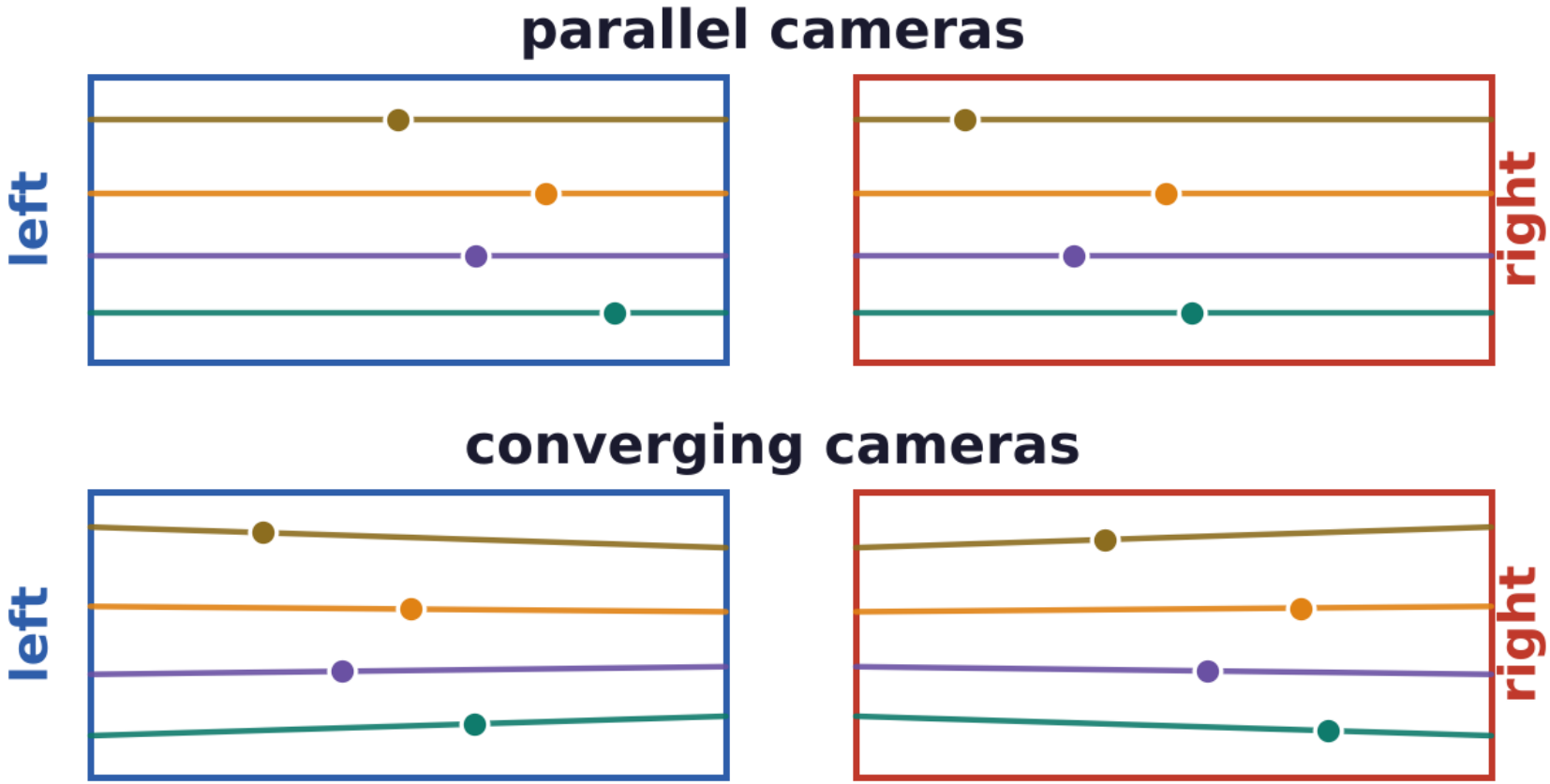


Figure 18: The same four scene points seen by a parallel pair of cameras (horizontal epipolar lines) and by a converging pair, each camera turned 12° inward (slanted lines). Each colored line is the epipolar line of one point; the dot is the point's image.

Camera calibration finds the relative positions of cameras

Camera calibration finds the relative positions and viewpoints of cameras, needed for binocular stereo, structure from motion, depth from multiple cameras and three-dimensional (3D) reconstruction, such as Gaussian splatting. The human visual system also calibrates itself. Calibration assumes corresponding points, found with interest points (distinctive points, such as corners). Theorems informally called "M views of N points" find the minimum number of points needed (Hartley and Zisserman, 2003; Ma et al., 2004).

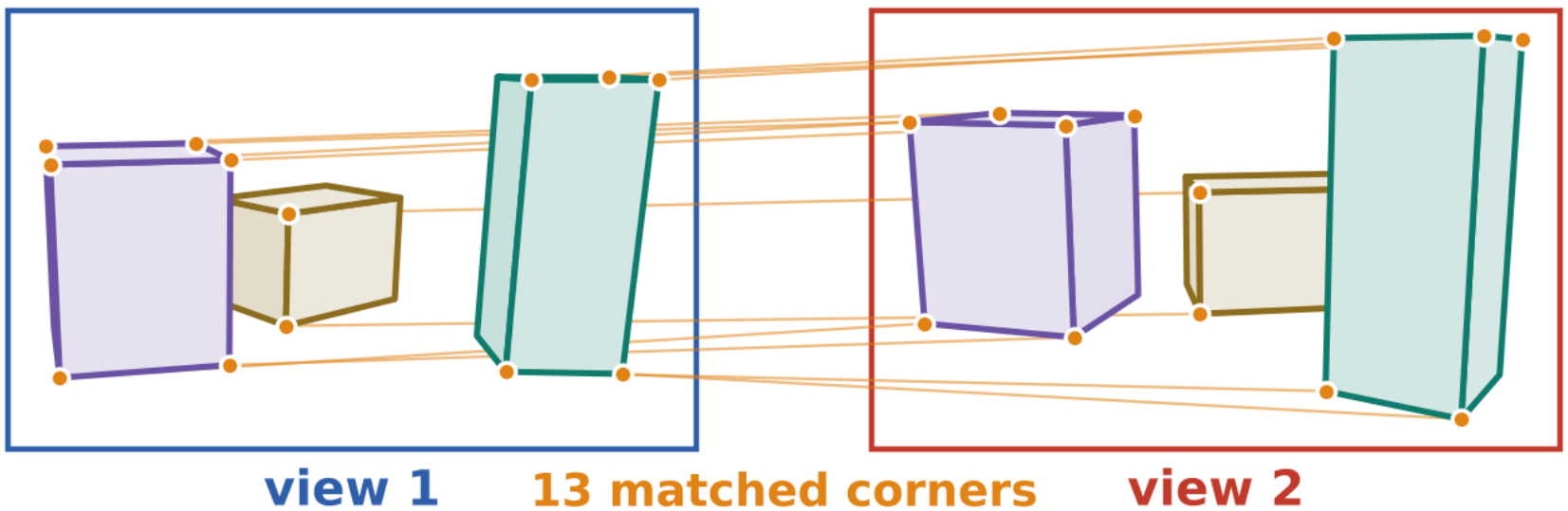


Figure 19: Three boxes in two views; the 13 corners visible in both, matched (orange).

Homogeneous coordinates: an image point is a ray

A point has coordinates $\vec{X} = (X_1, X_2, X_3)$ in the first camera's coordinate system (axes $\vec{a}, \vec{b}, \vec{c}$, origin at the pinhole) and $\vec{X}' = (X_1', X_2', X_3')$ in the second's; X_3 is the depth. As in stereo, the image plane is drawn in front of the pinhole, so the projections have no minus sign:

$$\begin{pmatrix} u \\ v \end{pmatrix} = \frac{f}{X_3} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}, \qquad \begin{pmatrix} u' \\ v' \end{pmatrix} = \frac{f}{X_3'} \begin{pmatrix} X_1' \\ X_2' \end{pmatrix}.$$

With u and v in units of the focal length ($f = 1$), $\vec{u} = (u, v, 1) = \vec{X}/X_3$ is the point divided by its depth. Every multiple of $(u, v, 1)$ lies on the same ray through the pinhole, so an image point is a vector defined up to scale: its homogeneous coordinates (Hartley and Zisserman, 2003).

A rotation R and a translation $\vec{t}$ relate the two cameras,

$$\vec{X}' = R\,(\vec{X} - \vec{t}\,),$$

so $\vec{t}$, where $\vec{X}' = 0$, is the second camera's center in first-camera coordinates.

Homogeneous coordinates: an image point is a ray

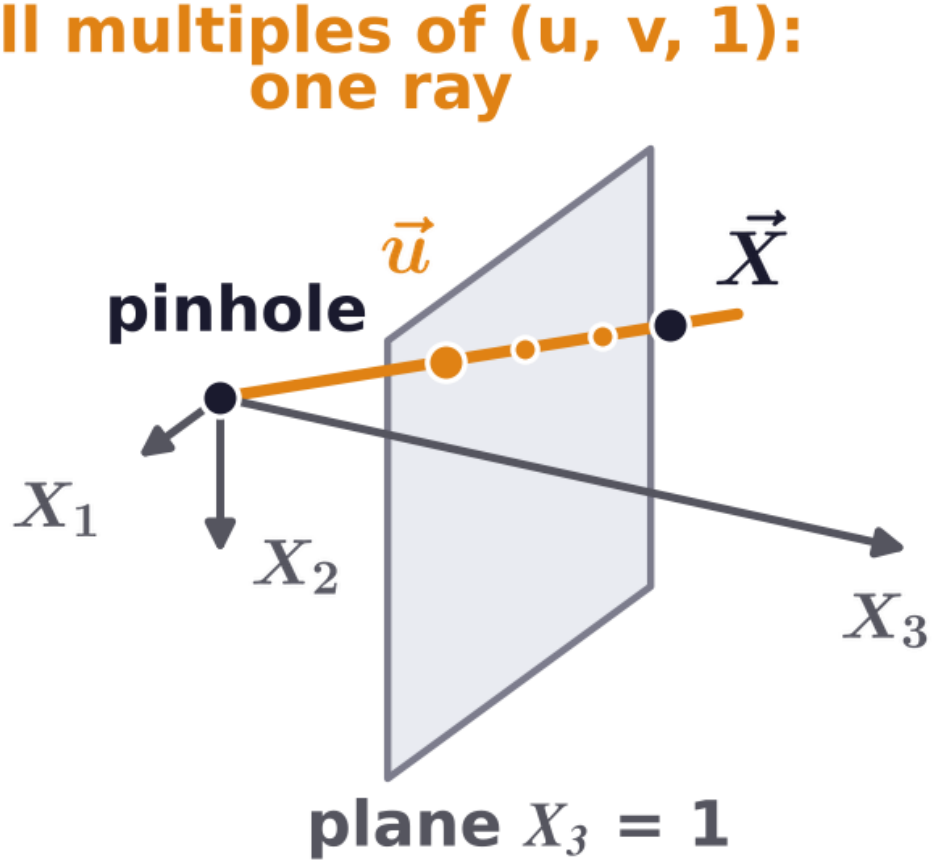


Figure 20: An image point in homogeneous coordinates: all multiples of $(u, v, 1)$ form one ray through the pinhole; it meets the plane $X_3 = 1$ at $\vec{u}$ and passes through the scene point $\vec{X}$.

The essential matrix combines the rotation and translation

The epipolar constraint uses the essential matrix

$$E = R\,[\vec{t}\,]_\times, \qquad ([\vec{t}\,]_\times)_{ij} = \sum_k \epsilon_{ijk}\, t_k, \qquad [\vec{t}\,]_\times = \begin{pmatrix} 0 & t_3 & -t_2 \\ -t_3 & 0 & t_1 \\ t_2 & -t_1 & 0 \end{pmatrix},$$

where ϵ_{ijk} is the anti-symmetric tensor: 0 if two indices are equal, 1 for even and -1 for odd permutations of $1, 2, 3$. Since the cross product is $(\vec{p} \times \vec{q})_i = \sum_{jk} \epsilon_{ijk} p_j q_k$,

$$([\vec{t}\,]_\times \vec{X})_i = \sum_{jk} \epsilon_{ijk} t_k X_j = (\vec{X} \times \vec{t}\,)_i :$$

multiplying by $[\vec{t}\,]_\times$ takes the cross product with $\vec{t}$, perpendicular to both $\vec{X}$ and $\vec{t}$. E holds the six camera parameters (three for R, three for $\vec{t}$).

Hartley and Zisserman (2003) write $\vec{X}' = R\vec{X} + \vec{t}\,'$ and $E = [\vec{t}\,']_\times R$ with $[\vec{t}\,']_\times \vec{X} = \vec{t}\,' \times \vec{X}$. Here $\vec{t}\,' = -R\vec{t}$; since $[R\vec{p}\,]_\times = R[\vec{p}\,]_\times R^T$ and their $[\vec{p}\,]_\times$ is minus ours, the signs cancel: E is the same.

The essential matrix combines the rotation and translation

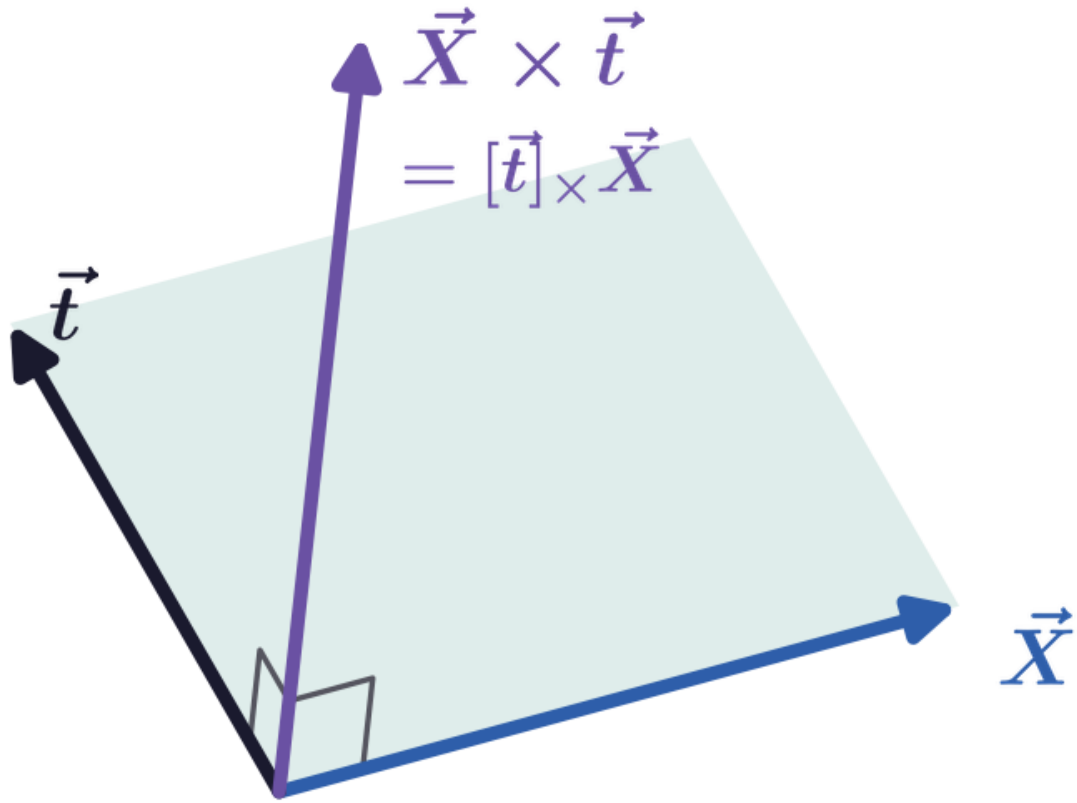


Figure 21: $\vec{X}$ (blue), $\vec{t}$ (black) and the cross product $\vec{X} \times \vec{t} = [\vec{t}]_\times \vec{X}$ (purple), perpendicular to the plane of $\vec{X}$ and $\vec{t}$.

Corresponding points satisfy the epipolar constraint

Corresponding points satisfy

$$\vec{u}'^{\,T} E\, \vec{u} = 0.$$

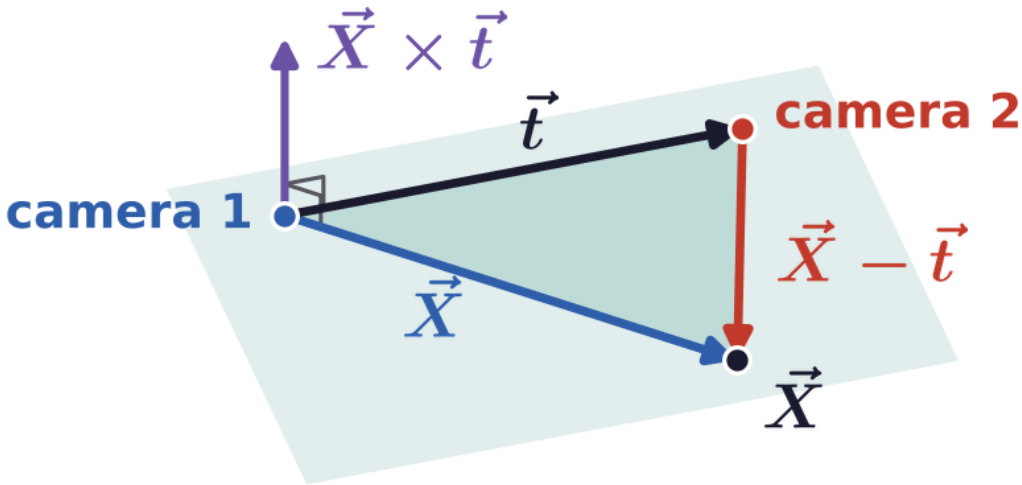
Proof. The vectors $\vec{u}$ and $\vec{u}'$ are proportional to $\vec{X}$ and $\vec{X}'$, so

$$\vec{u}'^{\,T} E\, \vec{u} \;\propto\; \vec{X}'^{\,T} E\, \vec{X} = (\vec{X} - \vec{t}\,)^T R^T R\, [\vec{t}\,]_\times \vec{X} = (\vec{X} - \vec{t}\,) \cdot (\vec{X} \times \vec{t}\,) = 0,$$

using $\vec{X}' = R(\vec{X} - \vec{t}\,)$; the rotation cancels because $R^T R = I$. The last product is zero because $\vec{X} \times \vec{t}$ is perpendicular to $\vec{X}$ and to $\vec{t}$, hence to $\vec{X} - \vec{t}$. Geometrically, $\vec{X}$, $\vec{X} - \vec{t}$ and $\vec{t}$ (the rays from the two camera centers to the point, and the baseline) lie in one plane, the epipolar plane, and $\vec{X} \times \vec{t}$ is its normal.

Each pair of corresponding points gives one equation for the camera parameters R and $\vec{t}$. For a fixed $\vec{u}$, write $\vec{l} = E\vec{u}$: the constraint $l_1 u' + l_2 v' + l_3 = 0$ is linear in (u', v'), so the points $\vec{u}'$ that satisfy it form a line, the epipolar line of $\vec{u}$ in the second image. On 200 random cameras and points, $|\vec{u}'^{\,T} E \vec{u}| \leq 1.4 \times 10^{-12}$.

Corresponding points satisfy the epipolar constraint



the epipolar plane through the two cameras and X; X × t is its normal

Figure 22: Camera 1 at the origin, camera 2 at $\vec{t}$: the vectors $\vec{X}$ (blue), $\vec{t}$ (black) and $\vec{X} - \vec{t}$ (red) lie in the epipolar plane (teal), and $\vec{X} \times \vec{t}$ (purple) is its normal.

Point matches determine R and the direction of t

The rotation R has three parameters and the translation $\vec{t}$ three. But replacing $\vec{t}$ by $s\vec{t}$ multiplies E by s and leaves the epipolar constraint unchanged: scaling the scene and the baseline together gives the same images. Only the direction of $\vec{t}$ can be found, so there are five unknowns, and the size of the scene is lost.

Each correspondence gives one equation. Five correspondences in general position (not aligned in any unlikely way) determine R and the direction of $\vec{t}$ up to a finite number of solutions, at most ten (Nistér, 2004).

The linear algorithm of Longuet-Higgins (1981) uses eight. The constraint $\sum_{ij} u'_i E_{ij} u_j = 0$ is linear in the nine entries of E, which are defined up to scale; eight correspondences give eight linear equations for their eight ratios. R and the direction of $\vec{t}$ are then extracted from E (Hartley and Zisserman, 2003).

Point matches determine R and the direction of t

In the figure a scene of three points is seen by two cameras, and the same scene with the points and the baseline doubled. In each camera the images of the two scenes coincide: the dots and the circles overlap.

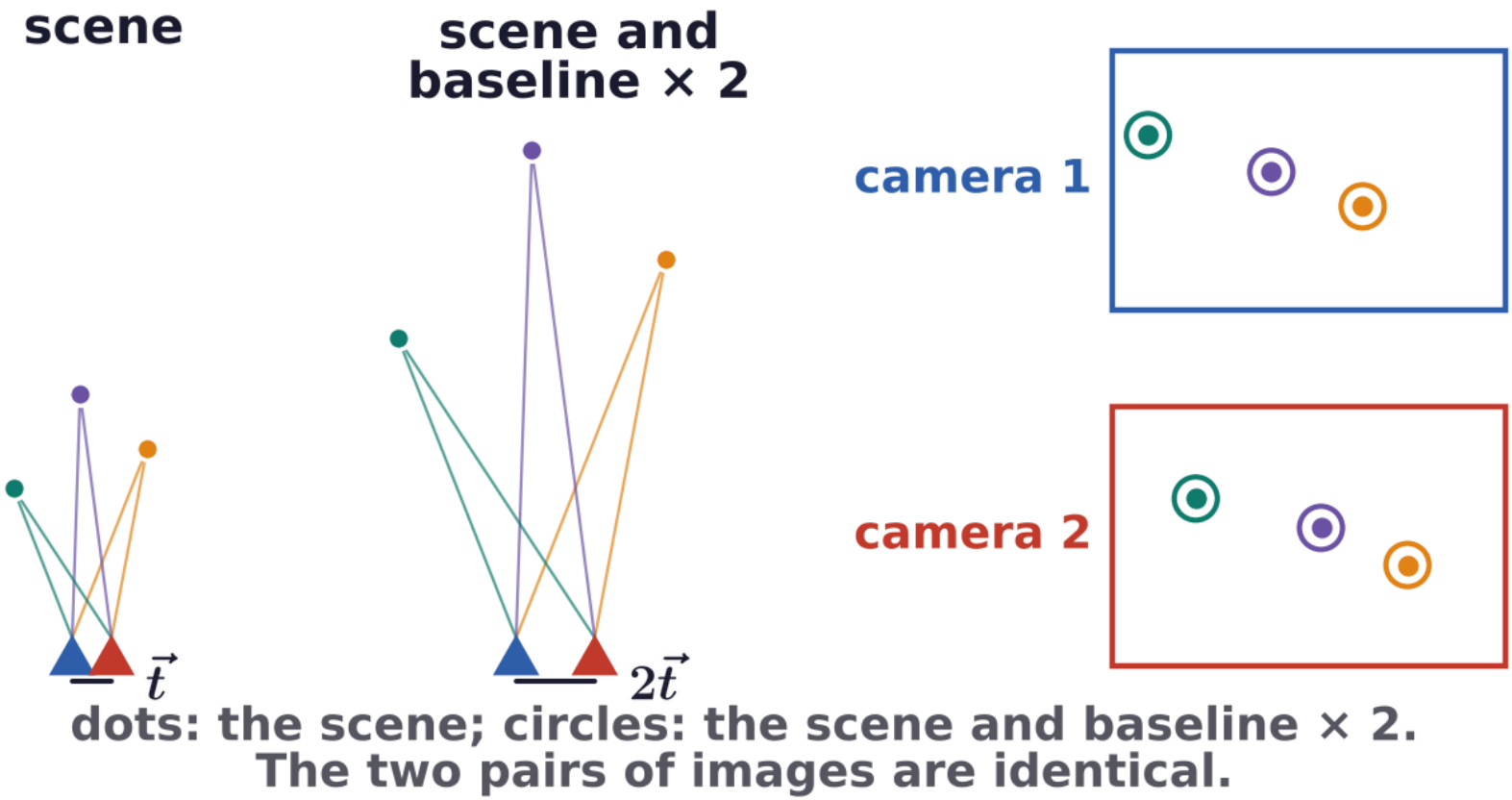


Figure 23: The two scenes from above (cameras: triangles), and the images in camera 1 and camera 2.

The fundamental matrix handles uncalibrated cameras

The essential matrix needs the normalized coordinates $\vec{u}$, in units of the focal length and measured from the image center, so the focal length and the image center must be known. A real camera reports pixel coordinates $\vec{p}$, related by $\vec{p} = K\vec{u}$, where the calibration matrix

$$K = \begin{pmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix}$$

holds the focal lengths in pixels and the image center (c_x, c_y); K' is the second camera's. Substituting $\vec{u} = K^{-1}\vec{p}$ and $\vec{u}\,' = K'^{-1}\vec{p}\,'$ into the epipolar constraint gives

$$\vec{p}\,'^{\,T} F\, \vec{p} = 0, \qquad F = K'^{-T} E\, K^{-1},$$

the fundamental matrix, where K'^{-T} is the transpose of K'^{-1}. As for E, the constraint is linear in the nine entries of F, defined up to scale, so eight matches find F without knowing K (Faugeras, 1992; Hartley, 1992; Luong and Faugeras, 1996). F has rank two, like $[\vec{t}\,]_\times$ ($[\vec{t}\,]_\times \vec{t} = 0$).

The fundamental matrix handles uncalibrated cameras

Zhang (2000) finds K itself from images of a planar pattern, such as a checkerboard, shown to the camera at a few (at least two) different orientations. Each view relates the plane to the image by a projective transformation (a 3×3 matrix acting on homogeneous coordinates), which constrains the entries of K; with several orientations of the pattern, K is determined.

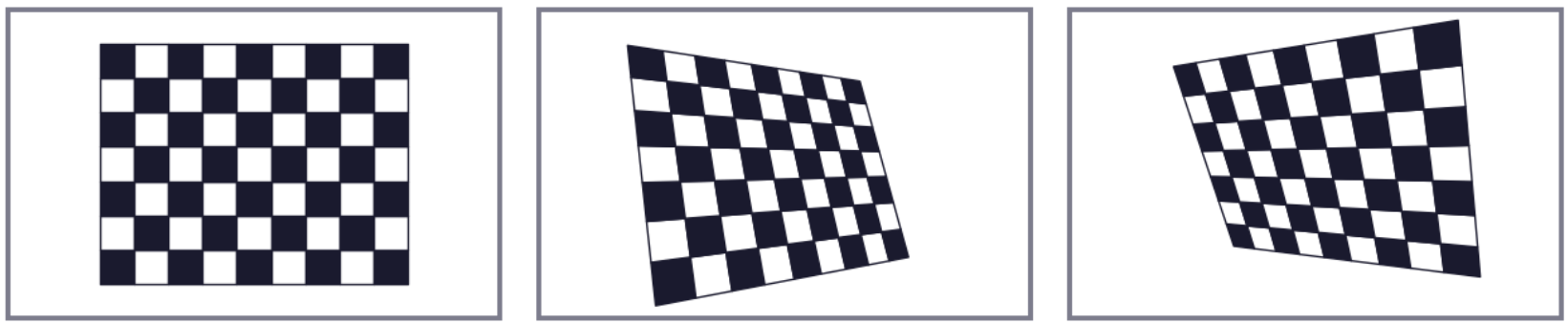


Figure 24: A 7×9 checkerboard seen by one camera at three orientations (computed perspective images).

Structure from motion estimates shape and viewpoints

Structure from rigid motion assumes a set of images, or views, of a rigid scene or object. We detect interest points in each image, correspond them between the images, calibrate the cameras (estimate the viewpoints) and estimate the shape. With linear projection the 3D structure, and the camera parameters up to an ambiguity, can be estimated by linear algebra: the singular value decomposition (SVD).

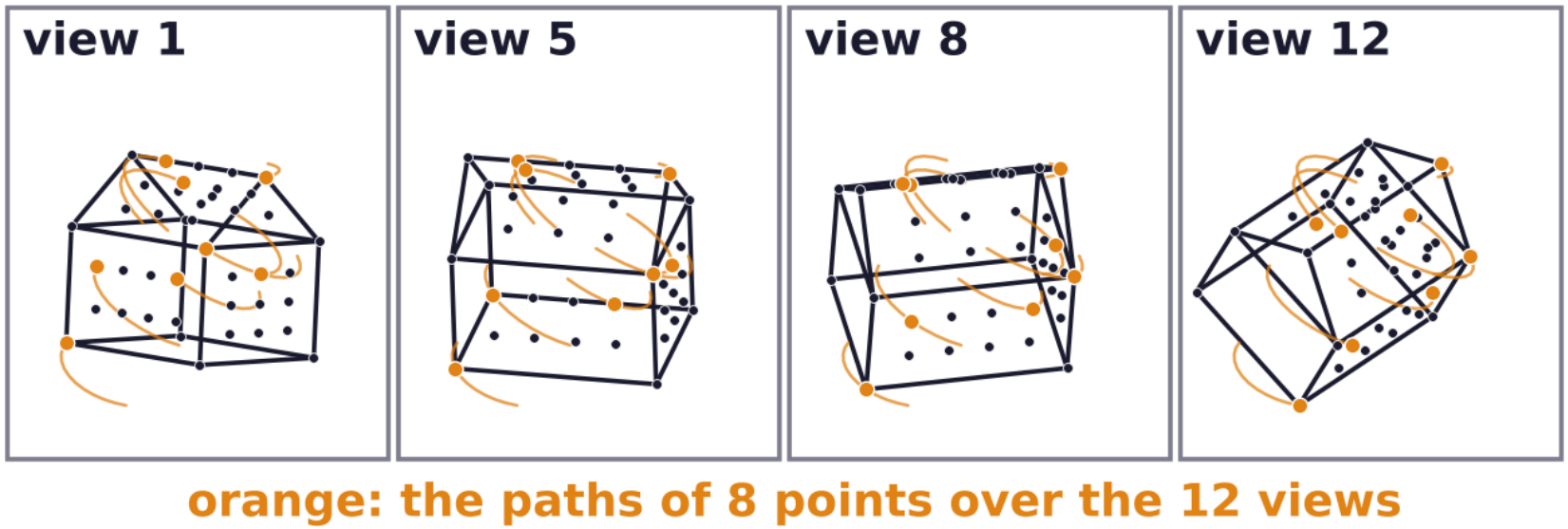


Figure 25: Four of the $M = 12$ views of a synthetic house of $N = 42$ points, turning by -55° to $+55^\circ$ in front of a scaled orthographic camera that looks down on it from 7° to 53° above; orange: the paths of eight points over all twelve views.

Under linear projection the measurements have rank three

Represent the object by N points $\vec{X}^n = (X_1^n, X_2^n, X_3^n)$, the interest points that can be detected and matched. View m projects with the rows $\vec{K}^m$ and $\vec{H}^m$ of an affine camera, $m = 1, \ldots, M$, and the observations are

$$u^{n,m} = \vec{K}^m \cdot \vec{X}^n = \sum_{i=1}^{3} K_i^m X_i^n, \qquad v^{n,m} = \vec{H}^m \cdot \vec{X}^n.$$

Image coordinates are measured from the centroid of the points in each view. A camera translation adds a constant to each view; the centroid of the image points is the image of the centroid of the object, so subtracting it removes the translation (Tomasi and Kanade, 1992). Otherwise the rank below would be four.

Collect the $u^{n,m}$ into an $M \times N$ matrix U, with one row per view. The formula says $U = K\,X$, where K (not the calibration matrix) is the $M \times 3$ matrix whose rows are the $\vec{K}^m$ and X is the $3 \times N$ matrix whose columns are the points: the shape. A product through three dimensions has rank at most three, and so does the matrix of the $v^{n,m}$.

Under linear projection the measurements have rank three

Rank three means that every row of U, the u coordinates of one view, is a combination of the three rows of X: the image points of an object lie in a 3D space (Ullman and Basri, 1991). The rank is the number of nonzero singular values of U (see the SVD). For the house without noise they are 7.23, 2.75, 0.52, then below 10^{-15}. With Gaussian noise of standard deviation 0.01 (1 % of the house) on every coordinate they are 7.25, 2.76, 0.53, then 0.084 and less: three large values.

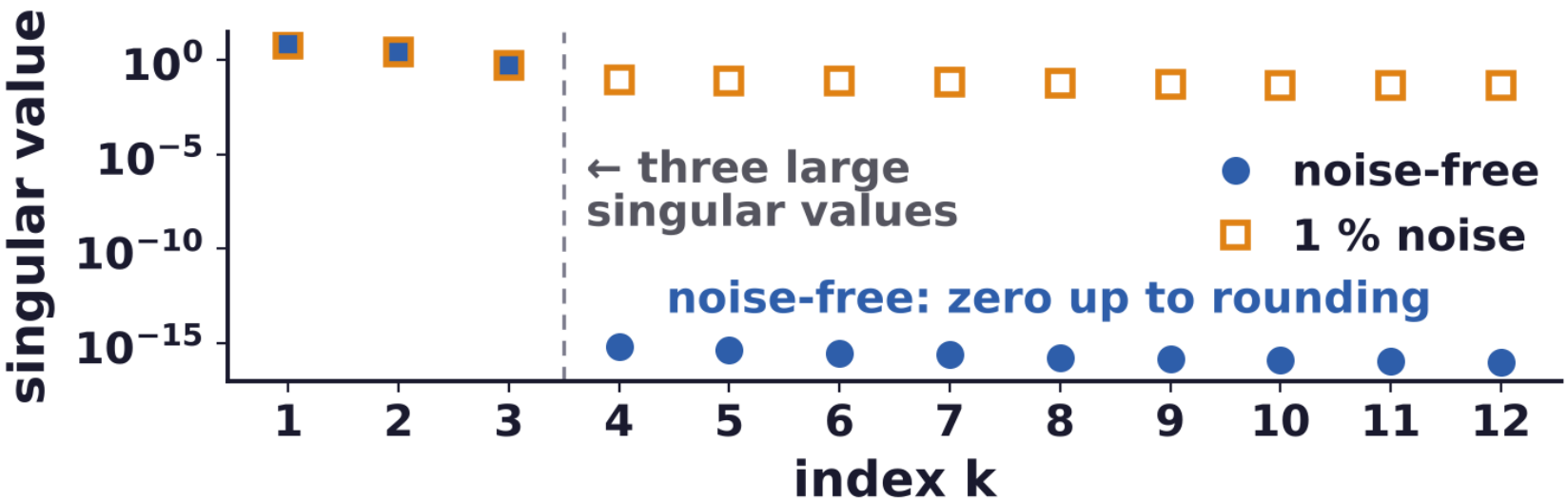


Figure 26: The singular values of the 12×42 matrix U of the house, without noise (blue dots) and with 1 % noise (orange squares).

A new view is a linear combination of known views

The u and the v coordinates of every view lie in the 3D space spanned by the three rows of the shape X. Two views give four rows, their u and their v, which in general span this space. So the coordinates of any new view are linear combinations of those in two model views:

$$u^{n,\mathrm{new}} = \sum_{k=1}^{4} c_k \, w_k^n, \qquad (w_1^n, w_2^n, w_3^n, w_4^n) = (u^{n,1}, v^{n,1}, u^{n,2}, v^{n,2}),$$

with four coefficients c_k for u and four for v, the same for all points. The four rows lie in a 3D space, so they are dependent and the c_k have many solutions (Ullman and Basri use three rows: $u^{n,1}$, $v^{n,1}$, $u^{n,2}$); every solution predicts the same view. Ullman and Basri (1991) used this for recognition: an object is recognized if a linear combination of its model views fits the image. The coefficients can be fitted from a few matched points and the fit checked on the rest; with translation, a constant is added to each combination.

A new view is a linear combination of known views

In the figure, view 7 of the house is predicted from views 1 and 12, the model views, by fitting the eight coefficients by least squares. With noise-free data the prediction (orange circles) agrees with the true view (dots) up to rounding: the largest error is 4×10^{-16}.

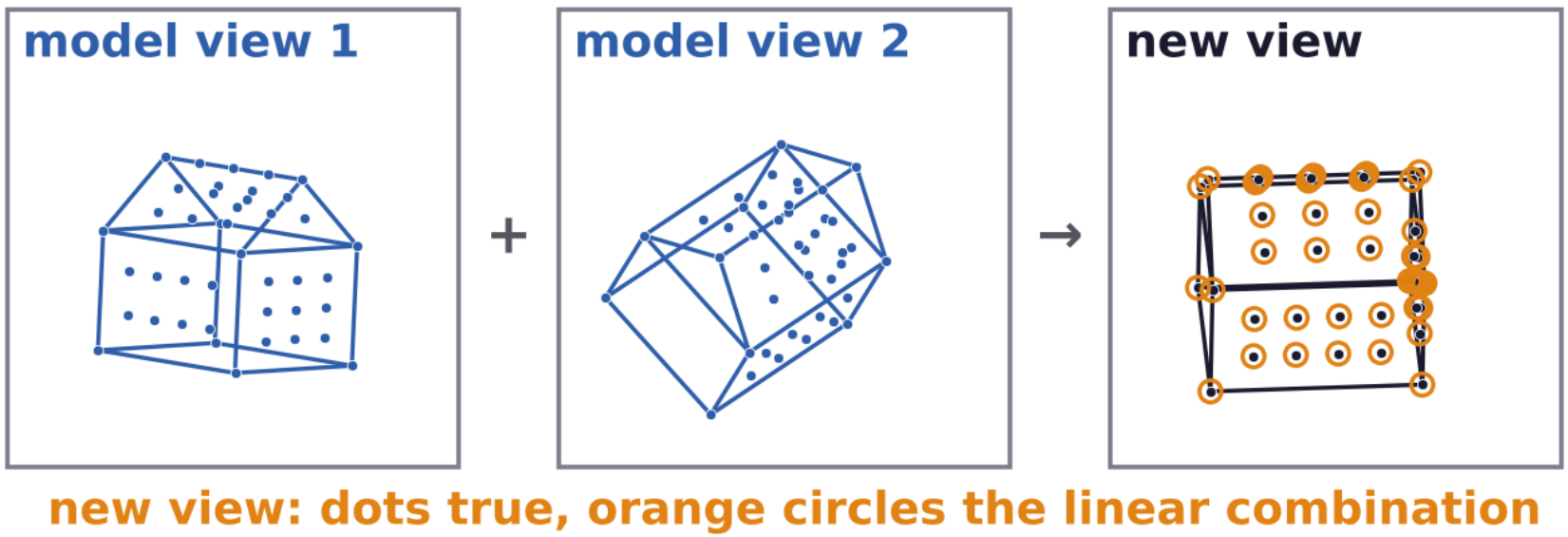


Figure 27: Model views 1 and 2 (views 1 and 12) and the new view (view 7): the linear combination of the model views (orange circles) on top of the true points.

Structure from motion as least squares: a bilinear problem

Estimate the shape $\{\vec{X}^n\}$ and the viewpoints $\{(\vec{K}^m, \vec{H}^m)\}$ from observations corrupted by additive zero-mean Gaussian noise of variance σ^2. The likelihood is $\propto e^{-E/2\sigma^2}$, where E, no longer the essential matrix, is the sum of squared errors

$$E[K, H, X] = \sum_{n,m} \left(u^{n,m} - \sum_{i=1}^{3} K_i^m X_i^n \right)^2 + \sum_{n,m} \left(v^{n,m} - \sum_{i=1}^{3} H_i^m X_i^n \right)^2,$$

so with a uniform prior the most probable shape and viewpoints minimize E. Rotating the object and the coordinates together leaves E unchanged.

Consider first the simpler case $E(K, X)$, with the u terms only. It is bilinear: for known K, E is quadratic in X and the best X solves linear equations, and vice versa. The SVD finds the global minimum; steepest descent also works: E is non-convex, but every local minimum is global; the other critical points are saddle points. The same landscape appears for a linear network that learns the principal components of its data (Baldi and Hornik, 1989).

Structure from motion as least squares: a bilinear problem

The figure shows steepest descent on the noisy house, from eight small random starts (step 0.008, 2500 steps). Every run first stalls on a plateau and then leaves it. The plateau at $E = 0.31$ is the rank-two saddle point: the best rank-two fit, whose error is $d_3^2 = 0.277$ ($d_3 = 0.526$, the third singular value of U) plus the minimum. Every run ends at the minimum the SVD gives, $E = 0.0368$, to within 10^{-3}.

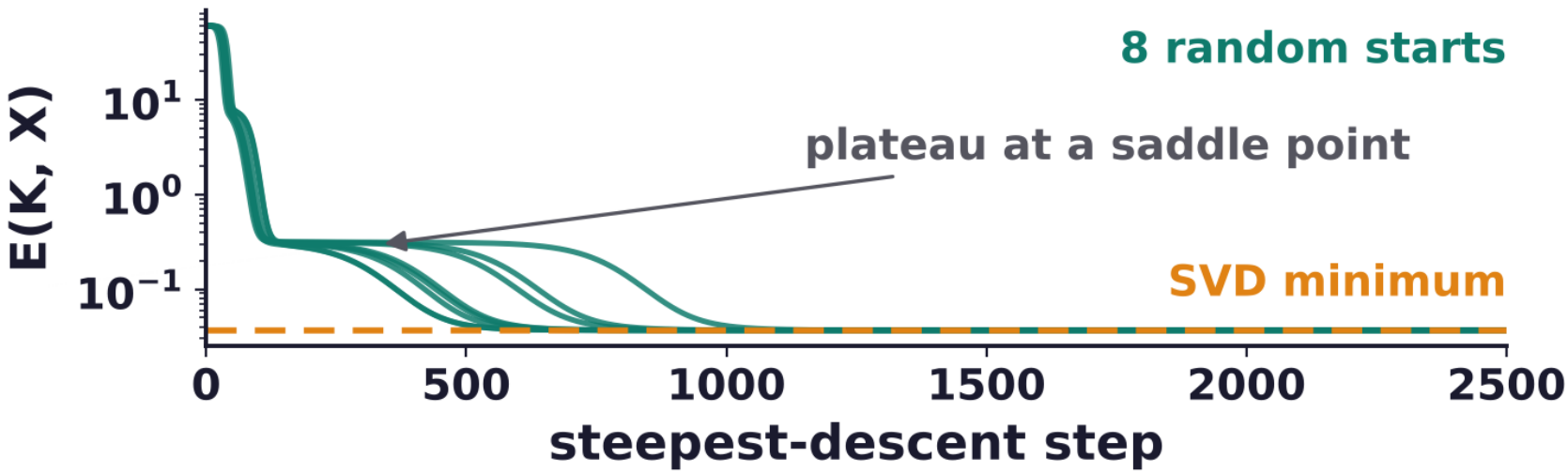


Figure 28: $E(K, X)$ during steepest descent from eight random starts (teal), and the SVD minimum (orange, dashed).

The singular value decomposition gives the global minimum

The SVD writes U as

$$U = E\,D\,F^T, \qquad D_{ij} = d_i\,\delta_{ij}, \qquad EE^T = I = FF^T,$$

where the letters E and F are reused for two orthogonal matrices. D is diagonal ($\delta_{ij} = 1$ if $i = j$, 0 otherwise), with the singular values in order, $d_1 \geq d_2 \geq d_3 \geq \cdots \geq 0$. The columns of E and F are the eigenvectors of UU^T and U^TU, with eigenvalues d_i^2.

Let $e_k(m)$ and $f_k(n)$ be the first three columns of E and F, and D_3 the top 3×3 block of D. The solutions are

$$K_i^m = \sum_{k=1}^{3} P_{ik}\,e_k(m), \qquad X_i^n = \sum_{k=1}^{3} Q_{ik}\,f_k(n), \qquad P^TQ = D_3,$$

for 3×3 matrices P and Q: then KX equals the first three terms of the SVD, $\sum_{k=1}^{3} d_k\,e_k(m)f_k(n)$. This is the best rank-three approximation of U (Eckart and Young, 1936), so it is the global minimum of the error $E(K, X)$, and the minimum value is $\sum_{k>3} d_k^2$.

The singular value decomposition gives the global minimum

In the figure the matrix U $(M \times N)$ is the product of E $(M \times M)$, D $(M \times N)$ and F^T $(N \times N)$. The rank-three fit keeps only the parts in orange: the first three columns of E, the top 3×3 block of D and the first three rows of F^T. For the noisy house the discarded part is $\sum_{k>3} d_k^2 = 0.0368$, the minimum that steepest descent reached.

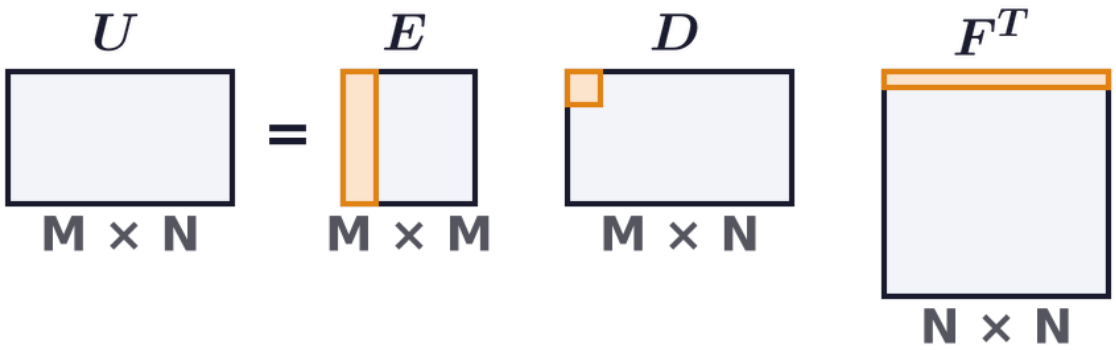


Figure 29: The shapes of the matrices in $U = EDF^T$; orange: the parts kept by the rank-three fit.

Orthogonality constraints leave an orthogonal ambiguity

Write K for the $M \times 3$ matrix of rows $\vec{K}^m$ and X for the $3 \times N$ matrix of points. For any invertible 3×3 matrix A,

$$K \mapsto KA, \qquad X \mapsto A^{-1}X$$

leaves KX, and so the error E, unchanged. Part of this ambiguity rotates the object and the coordinates; the rest distorts the shape. The u and v terms share the shape X, so stack their rows into one $2M \times N$ matrix: a single SVD gives rows $\hat{K}^m$, $\hat{H}^m$ and a shape $\hat{X}$, and the true rows are $\hat{K}^m A$, $\hat{H}^m A$ for one common A (Tomasi and Kanade, 1992). Then impose, for every view m, the conditions of scaled orthographic projection: $\vec{K}^m$ and $\vec{H}^m$ are orthogonal and of equal length,

$$\vec{K}^m \cdot \vec{H}^m = 0, \qquad |\vec{K}^m| = |\vec{H}^m|.$$

With $Q = AA^T$ (a new Q) these are linear equations in the six entries of the symmetric matrix Q: $\hat{K}^m Q \hat{H}^{mT} = 0$ and $\hat{K}^m Q \hat{K}^{mT} = \hat{H}^m Q \hat{H}^{mT}$. Their solution gives Q up to scale, and A follows from $Q = AA^T$ (from the eigenvectors of Q).

Orthogonality constraints leave an orthogonal ambiguity

$Q = AA^T$ is unchanged when A is replaced by AO with O orthogonal, so A, and the shape, are determined up to a scale and a rotation, possibly with a mirror reflection in depth (orthographic projection cannot tell a shape from its mirror image). In the figure the SVD shape with $A = I$ is an affine distortion of the house; after the constraints the recovered points, aligned by the best rotation, reflection and scale, fall on the true ones: the root-mean-square error is 0.7 % of the house size, against 5.5 % before.

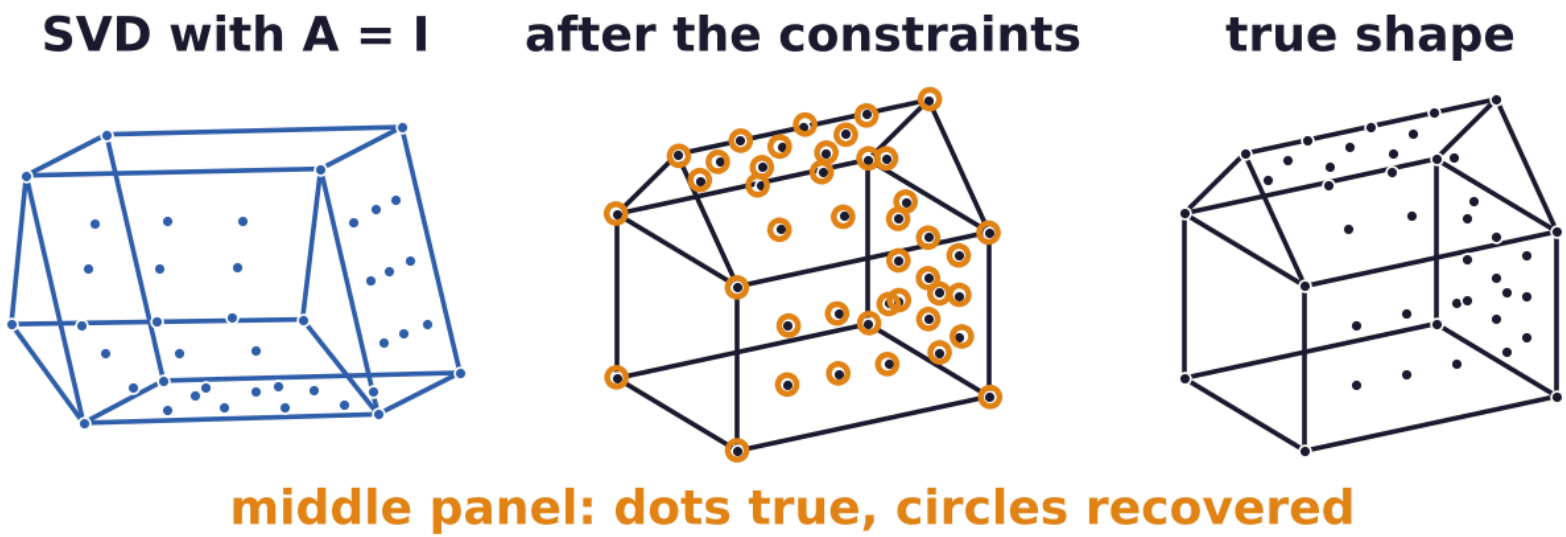


Figure 30: The SVD shape with $A = I$ (blue); after the constraints (orange circles); the true shape.

Factorization recovers shape from real and synthetic views

This algorithm, factorization, was first found by Kontsevich et al. (1987) and reinvented by Tomasi and Kanade (1992), who applied it to a real house sequence. It can be extended to perspective projection, but this requires a more complex algorithm.

The synthetic house recovered from its twelve noisy views agrees with the true shape up to rotation, scale and reflection.

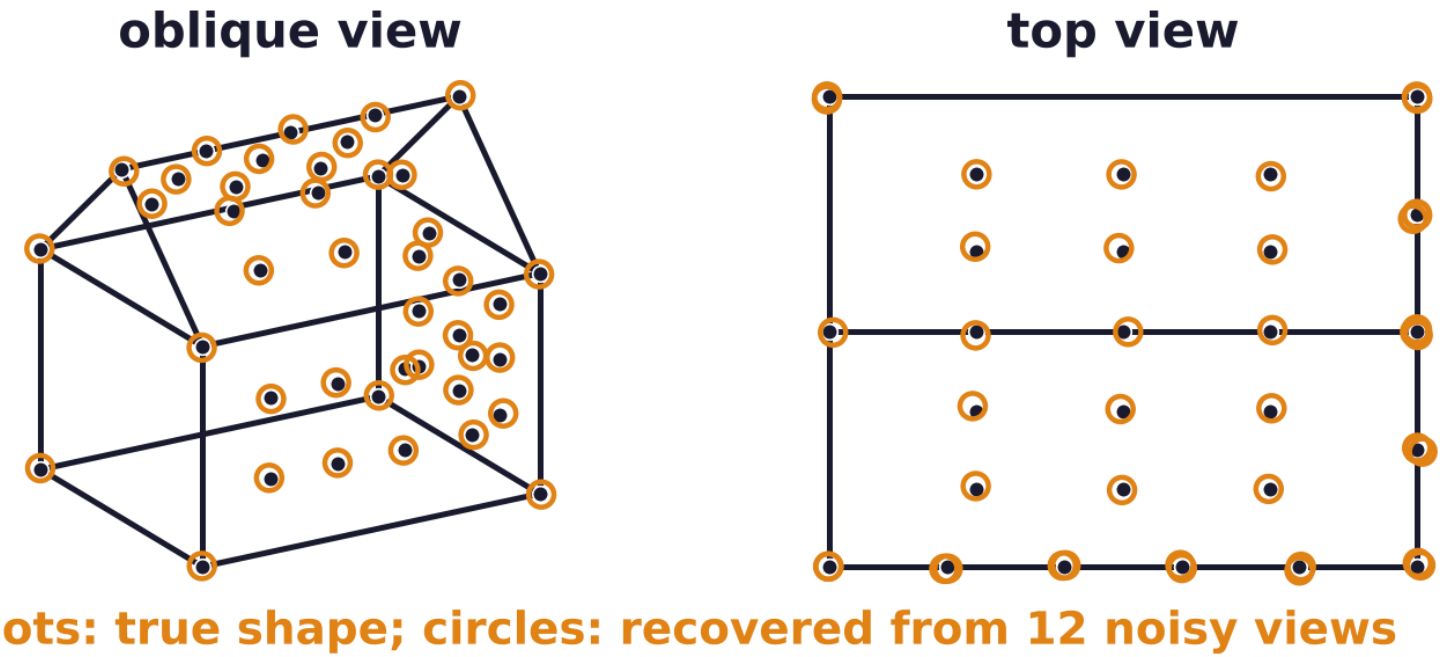


Figure 31: The synthetic house, true (dots) and recovered from twelve noisy views (orange circles), seen obliquely and from above.

The house sequence of Tomasi and Kanade

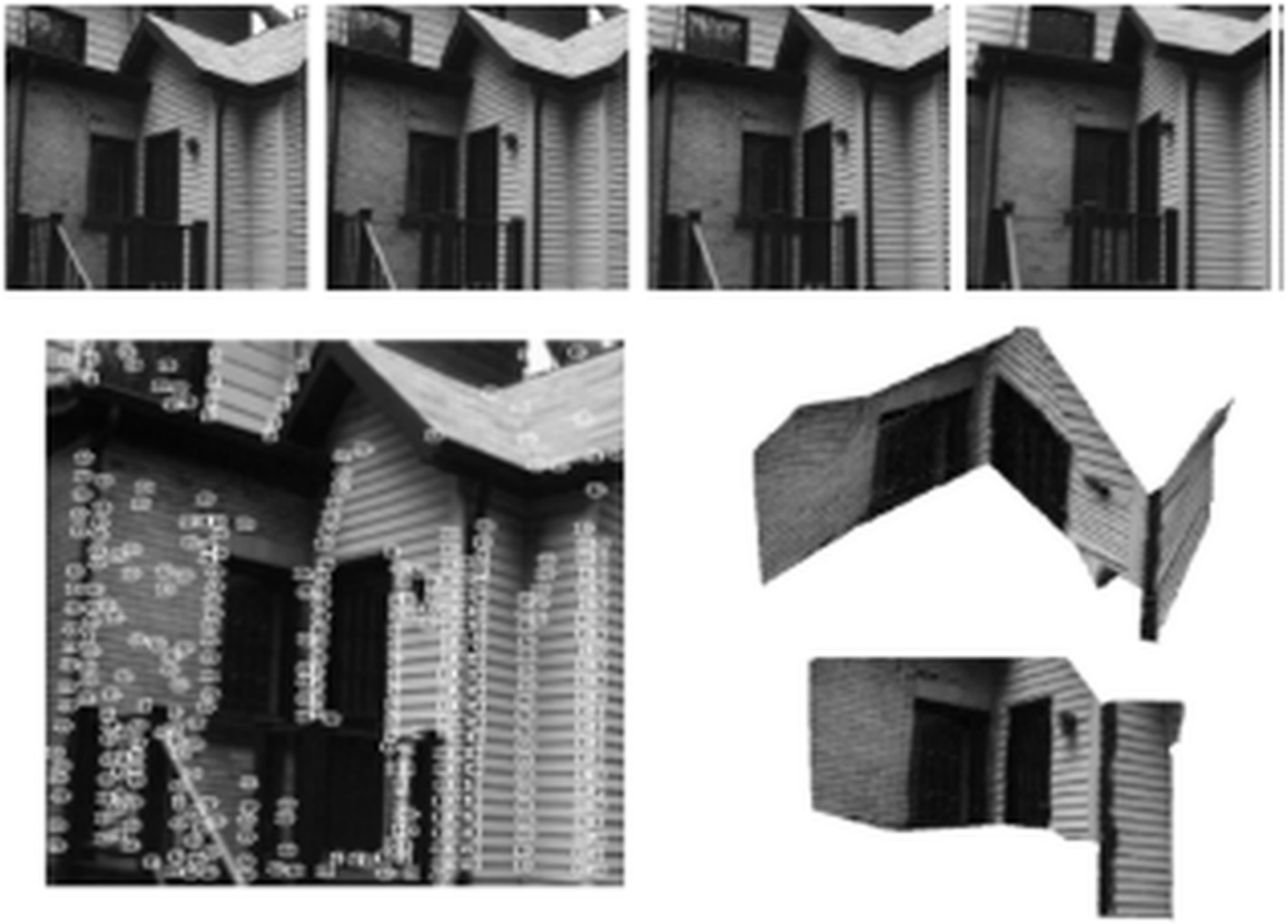


Figure 32: Four frames of a house sequence, the tracked features and the recovered shape with the image pasted on (Tomasi and Kanade, 1992; 2020 slides, slide 11). The picture is a low-resolution copy.

Non-rigid shapes as linear sums of basis shapes

The method extends to a special class of non-rigid motion: objects expressed as a linear sum of d basis shapes $\vec{b}_i$, with coefficients α_i^m that vary with time, the frame m,

$$\vec{X}^{n,m} = \sum_{i=1}^{d} \alpha_i^m \, \vec{b}_i^{\,n}, \qquad u^{n,m} = \sum_{i=1}^{d} \alpha_i^m \, \vec{K}^m \cdot \vec{b}_i^{\,n},$$

where $\vec{X}^{n,m}$ is point n in frame m and $\vec{b}_i^{\,n}$ is point n of basis shape i. The second formula is the projection of the first. It says $U = \tilde{K}\,\tilde{B}$, where row m of $\tilde{K}$ is $(\alpha_1^m \vec{K}^m, \ldots, \alpha_d^m \vec{K}^m)$, with $3d$ entries, and $\tilde{B}$ stacks the d basis shapes into $3d$ rows. So the measurements have rank at most $3d$, as Bregler et al. (2000) showed.

In the figure a face of 26 points is the sum of $d = 2$ basis shapes: the neutral face, and a deformation that opens the mouth, drops the jaw and raises the brows. Over 12 frames the face turns by $-35°$ to $+35°$ while the mouth opens and closes. U has six nonzero singular values ($3d = 6$; the sixth is small, 3×10^{-4}), and the rest are below 10^{-15}.

Non-rigid shapes as linear sums of basis shapes

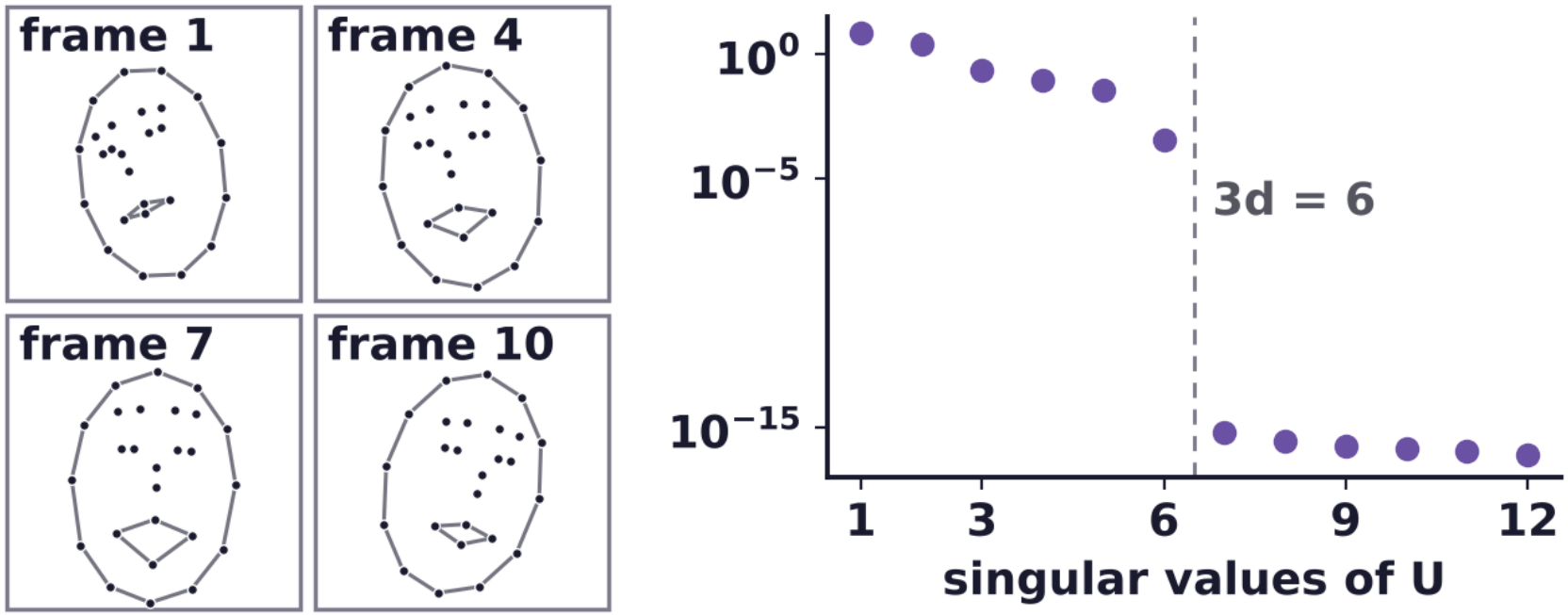


Figure 33: Four of the twelve frames of the face, and the singular values of its 12×26 matrix U: six nonzero ($3d = 6$), the rest zero up to rounding.

The non-rigid ambiguities are only coordinate changes

Initially the approach was poorly understood: researchers added unnecessary constraints to remove ambiguities in $\vec{b}$, α, $\vec{K}$ and $\vec{H}$. For example, any invertible $d \times d$ matrix G replaces the basis $(\vec{b}_1, \ldots, \vec{b}_d)$ by $(\vec{b}_1, \ldots, \vec{b}_d)G$ and the coefficients by $G^{-1}\alpha$, with the same shapes in every frame. It was realized that these are only coordinate changes: they leave the 3D shapes unchanged, so they need not be removed. This led to theoretically very attractive algorithms (Dai et al., 2012). But they are complex, are not guaranteed to reach the global minimum, and apply to limited types of motion: they cannot deal with articulated objects (rigid parts joined at joints) like humans and animals.

The non-rigid ambiguities are only coordinate changes

In the figure, basis 1 is $(\vec{b}_1, \vec{b}_2)$ (blue) and basis 2 is $(\vec{b}_1 + \vec{b}_2, \vec{b}_1 - \vec{b}_2)$ (orange), that is G with rows $(1, 1)$ and $(1, -1)$, with the coefficients changed to match. The frames generated from either basis are identical.

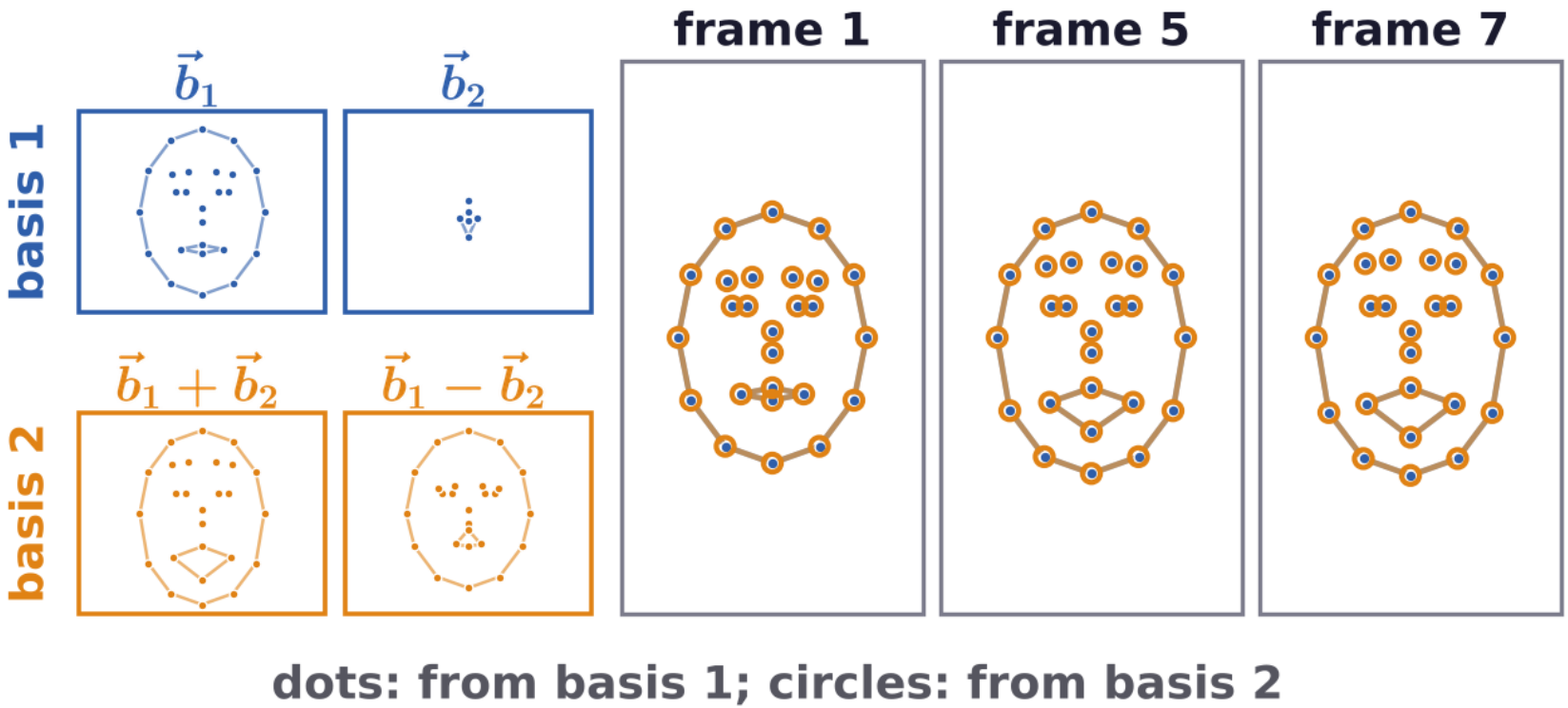


Figure 34: Two bases for the face and three frames generated from each: the dots and the circles coincide.

Painters exploited vanishing points to give depth

Geometric priors are assumptions about the scene's geometry that let one image give depth. European painters used vanishing points to give depth to their pictures, starting from crude attempts by Giotto. In the construction of a tiled floor by Alberti (1435) the lines into depth, the orthogonals (purple), meet at one point. In computer vision they give 3D structure and the ground plane, where trees and people stand vertically.

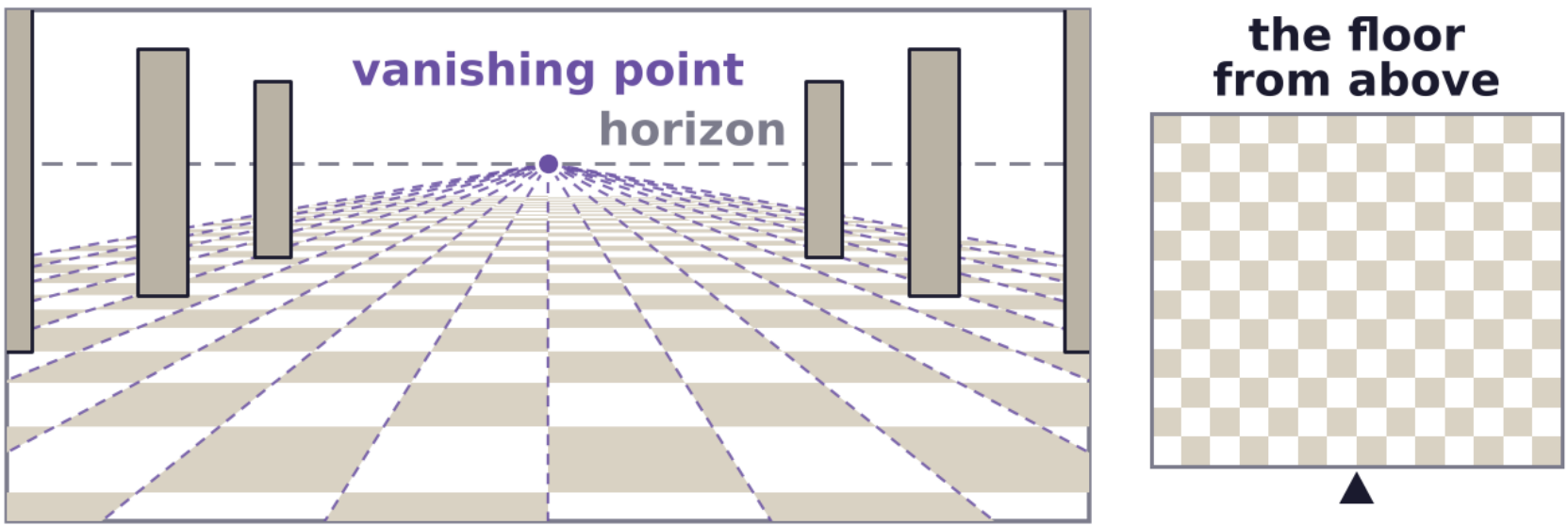


Figure 35: 1 m tiles and columns seen from 1.6 m; the floor from above (triangle: camera).

The Ames room breaks the assumptions of perspective

Humans use these cues to obtain depth from perspective, but the cues make strong assumptions about the viewed scene. If the assumptions are invalid, human perception makes mistakes. The Ames room looks like a real room from its viewpoint, a peephole, but it has a different geometry (Ittelson, 1952). It is built from a rectangular room by moving each corner along its own viewing ray from the peephole, so every corner, and every edge, projects exactly as before: the image is unchanged.

The viewer assumes the room is rectangular and the far wall at one depth. This fools humans: people seem to change in size when they move across the room.

The Ames room breaks the assumptions of perspective

In the figure the four left corners are moved to 1.9 times their distance from the peephole, so the walls, floor and ceiling stay planar while the far wall recedes toward the left side. Two people 1.7 m tall against the far wall are at depths 9.05 m and 5.45 m along the gaze; through the peephole the farther one looks 0.60 times as tall.

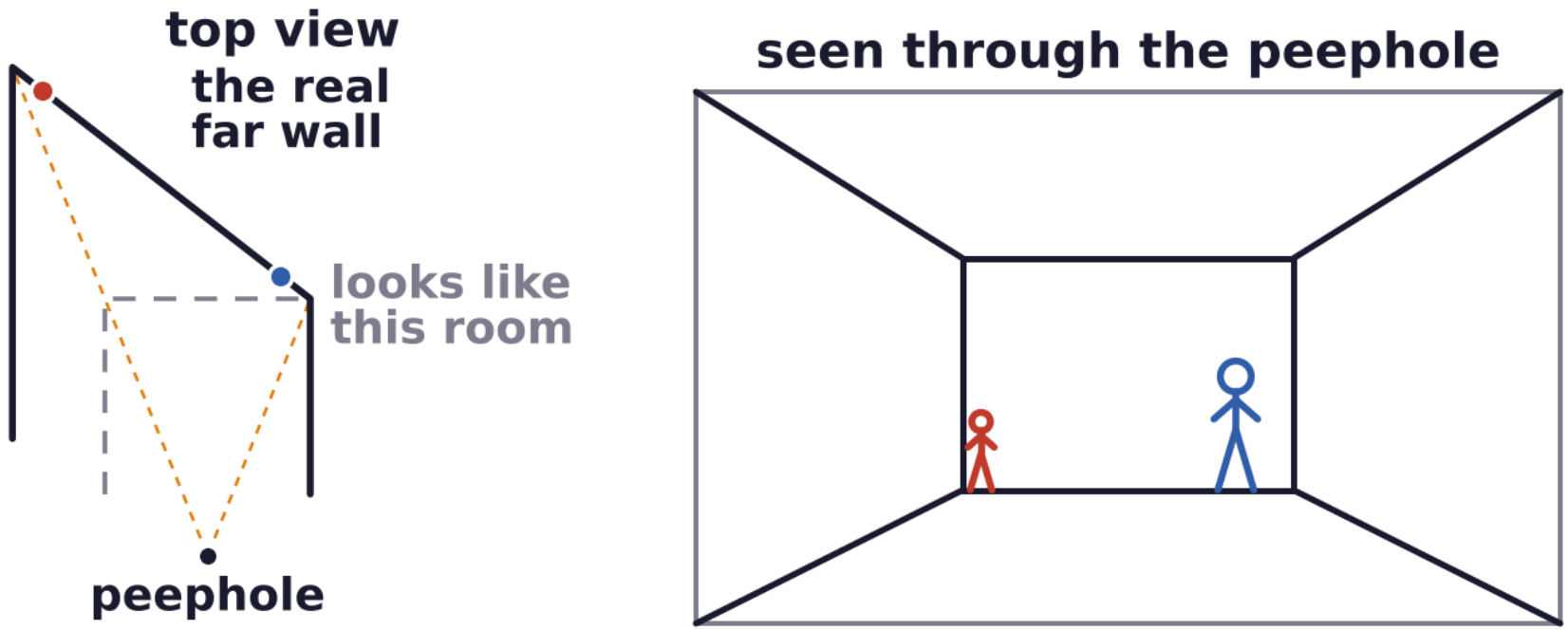


Figure 36: Top view: the rectangular room it looks like (dashed) and the real Ames room (black); orange: the rays to the far corners; red, blue: the two people. Second panel: the view through the peephole.

Manhattan World: man-made scenes have three main axes

Man-made scenes often have three dominant, perpendicular axes: Manhattan World. It gives the orientation of the camera with respect to this natural coordinate system of the scene, either from the vanishing points (an image of Manhattan can have three) or from the statistics of the edge orientations, if most edges come from the three axes (Coughlan and Yuille, 2003).

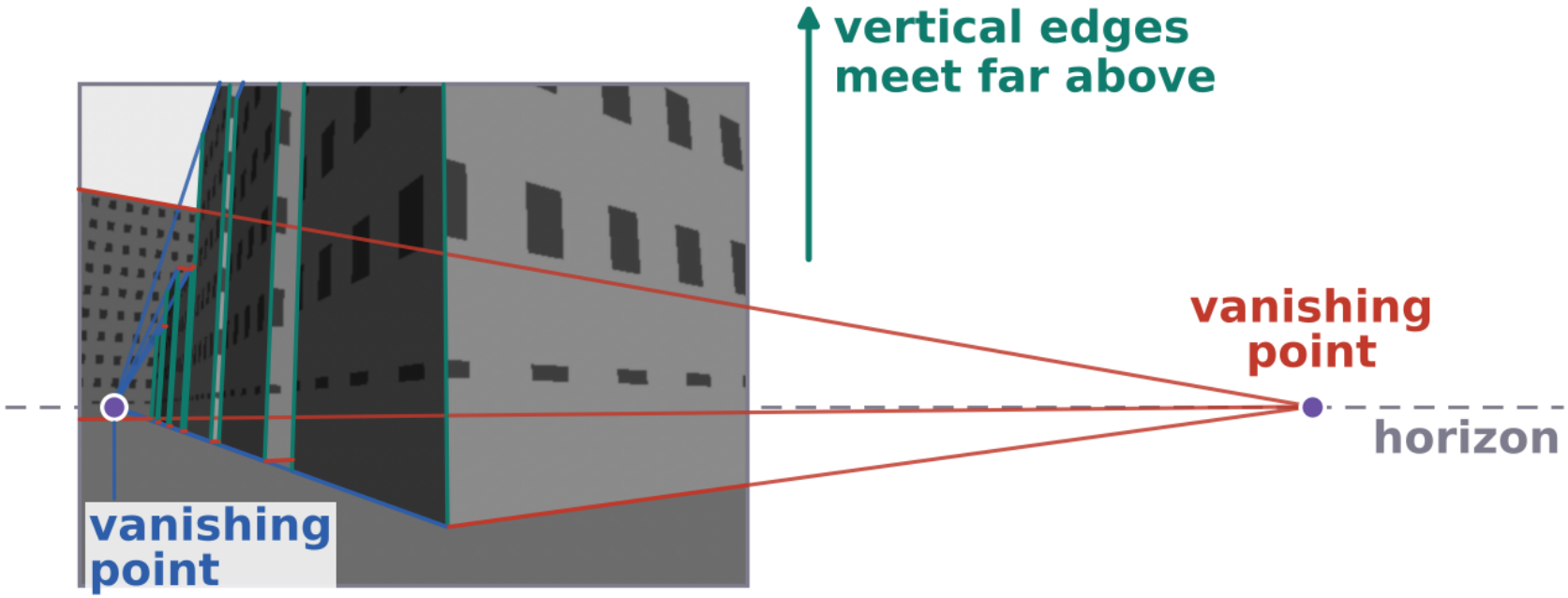


Figure 37: A synthetic street, edges colored by axis: two vanishing points on the horizon; the vertical one is far above.

Bayesian inference finds the compass angle from edges

Coughlan and Yuille (1999) estimate the compass angle Ψ, the camera's rotation about the vertical relative to the Manhattan axes, by Bayesian inference over all pixels, with no edge map. At position u the data are $\vec{E}_u = (E_u, \phi_u)$: the gradient magnitude (edge strength) and direction. A hidden variable m_u says what made the pixel: an edge of axis 1, 2 or 3 (the vertical), a random edge, or no edge, with prior $P(m_u) = 0.02, 0.02, 0.02, 0.04, 0.9$. With pixels independent given Ψ, Bayes' rule gives

$$P(\Psi \mid \{\vec{E}_u\}) \;\propto\; P(\Psi) \prod_u \sum_{m_u=1}^{5} P(E_u \mid m_u)\, P(\phi_u \mid m_u, \Psi, u)\, P(m_u).$$

$P(E_u \mid m_u)$ is the edge-strength distribution on edges (P_{on}, for $m_u \leq 4$) or off edges (P_{off}), learned from labeled images (Konishi et al., 2003). For $m_u \leq 3$ the edge, perpendicular to the gradient, must point toward axis m_u's vanishing point, which moves with Ψ: $P(\phi_u \mid m_u, \Psi, u)$ is $(1-\epsilon)/(4\tau)$ within τ of the predicted gradient direction or its opposite, and $\epsilon/(2\pi - 4\tau)$ elsewhere ($\epsilon = 0.1$, $\tau = 4^\circ$ to 6°); for random and no edges it is uniform.

Bayesian inference finds the compass angle from edges

With a uniform prior $P(\Psi)$ the posterior is a product over pixels of sums over m_u, searched over $-45° \leq \Psi \leq 45°$ (axes 1 and 2 are interchangeable). On 23 of 25 indoor images the error was below 5°.

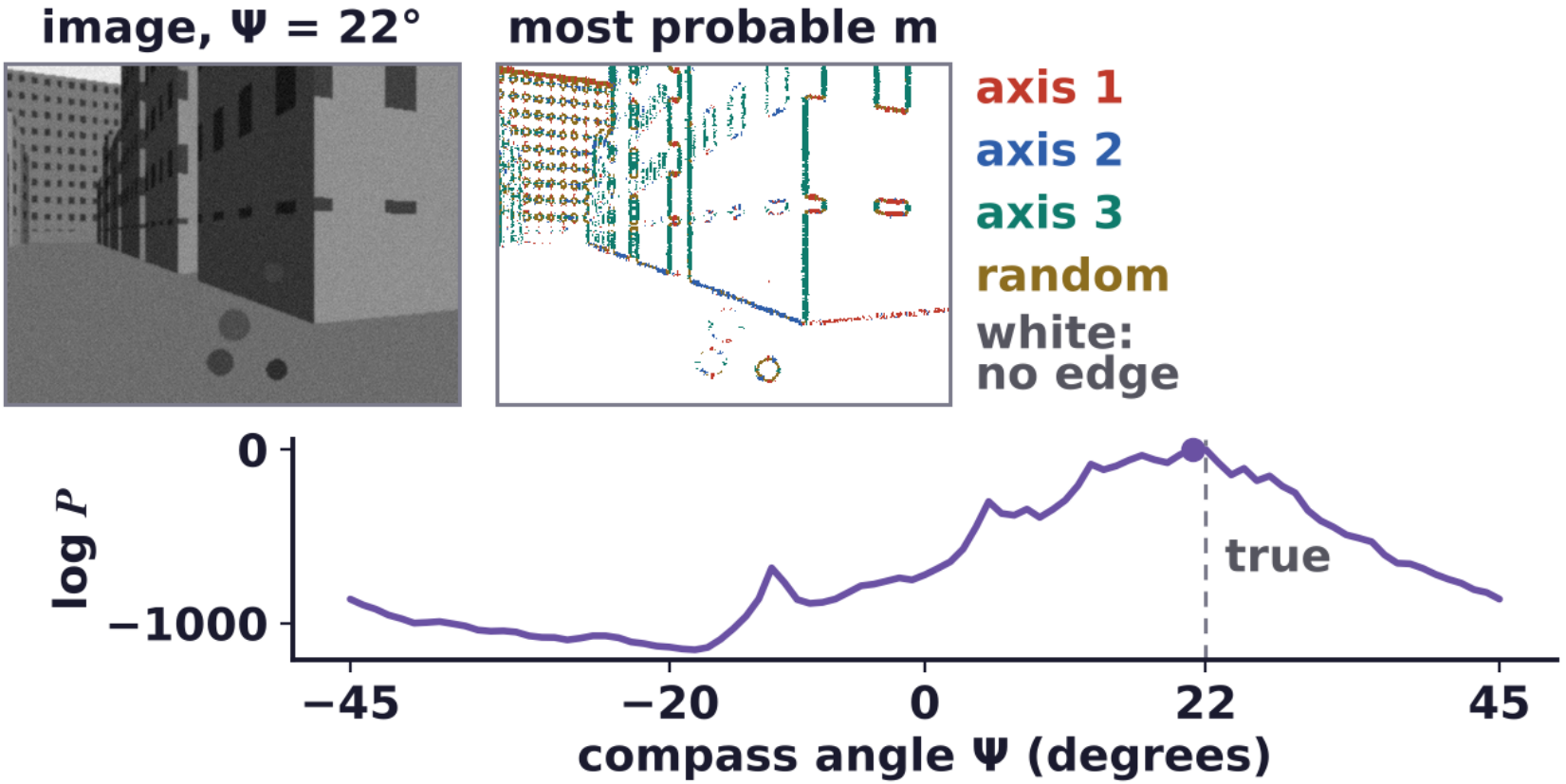


Figure 38: A synthetic street ($\Psi = 22°$), the most probable m_u, and the log posterior, with its peak at 21°.

Planes aligned to the three axes are a prior for stereo

The idea of Manhattan World has been extended to say that the world often consists of planar surfaces aligned to the three major axes, and this has been used as a prior for stereo; Furukawa et al. (2009) used it with many views. Every normal is then one of three directions, and depth is piecewise planar.

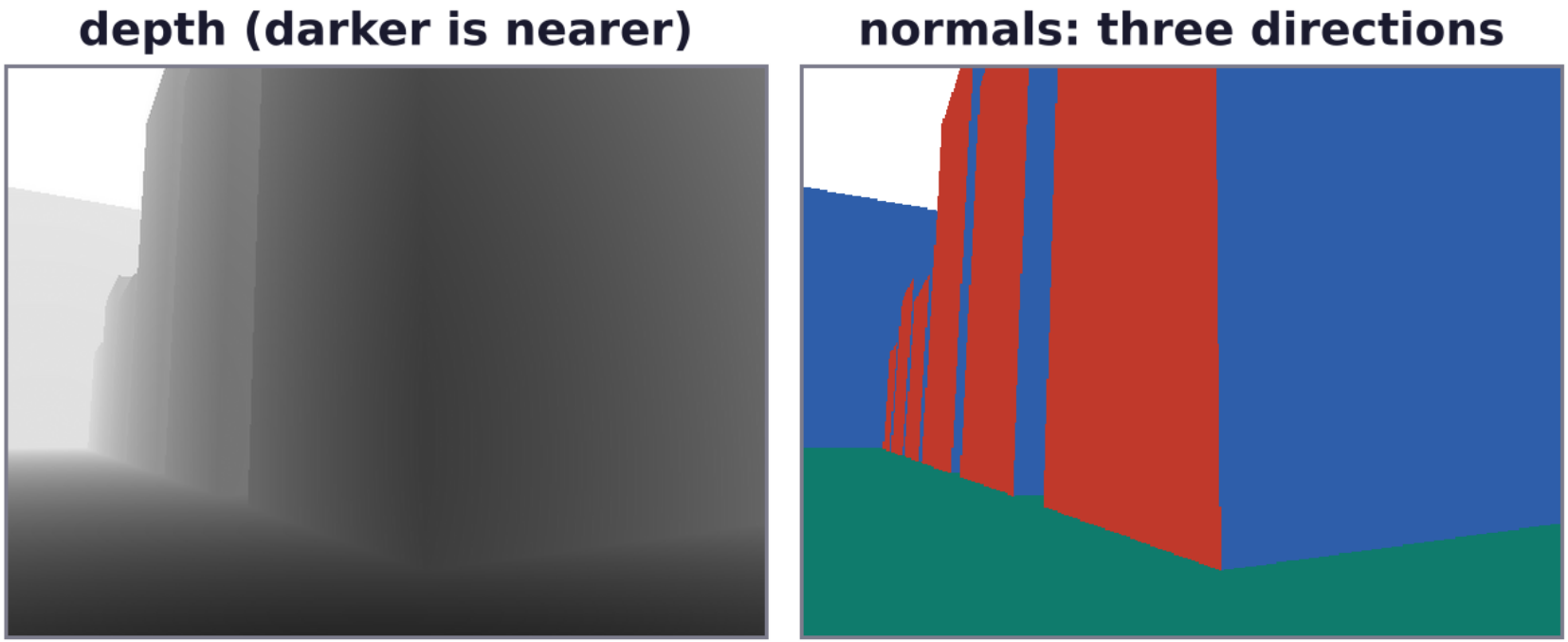


Figure 39: Depth (darker is nearer) and normals of the street, colored by axis (red 1, blue 2, teal vertical): a wall with red edges has a blue normal.

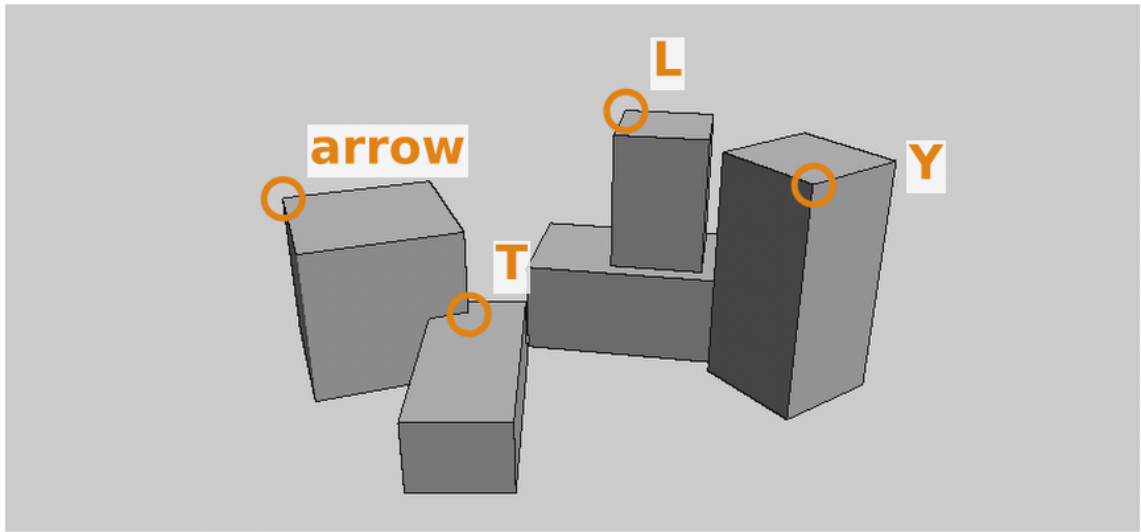
Structured scenes reveal their 3D geometry in a single image

This relates to the classic blocks world, constructed by placing a set of blocks on the ground (Roberts, 1963). It was shown that analyzing the boundaries of the blocks determined their three-dimensional structure. The boundaries meet at junctions: L (one face's corner), arrow and Y (two or three visible faces at a corner) and T (an edge passing behind a block). Each junction allows only a few 3D readings of its edges, as convex, concave or occluding (one surface hides another), and junctions that share an edge must agree (Huffman, 1971; Clowes, 1971).

This was extended to more realistic situations in cities. More generally, if the world is generated by a grammar (rules that build scenes from parts) regular enough, inverting the grammar can give the 3D structure of the scene. If the viewed scene is structured enough, much of its geometry can be estimated from a single image.

Structured scenes reveal their 3D geometry in a single image

In the figure four blocks stand on the ground and a fifth on top of another, rendered with hidden surfaces removed. One junction of each kind is circled; the key below says what meets there.



Y	**three faces meet at a corner**
arrow	**two faces, an outer corner**
L	**one face's corner**
T	**an edge passes behind a block**

Figure 40: The blocks world: L, arrow, Y and T junctions.

Summary

- Pinhole camera: $u = -f\,(\vec{r}\cdot\vec{a})/(\vec{r}\cdot\vec{c})$. Perspective keeps lines straight; dividing by depth makes parallel lines converge.
- All lines with one direction share a vanishing point, unless they are perpendicular to the gaze.
- Scaled orthographic and affine cameras approximate projection linearly, and lose the vanishing points.
- Stereo needs calibration; corresponding points lie on corresponding epipolar lines.
- Essential matrix: $\vec{u}'^{\,T} E\,\vec{u} = 0$ gives R and the direction of $\vec{t}$; the fundamental matrix F works in pixels.
- Rigid structure from motion: the measurements have rank three; the SVD gives shape and viewpoints.
- Non-rigid motion as sums of d basis shapes has rank at most $3d$; its ambiguities are coordinate changes.
- Geometric priors (ground plane, Manhattan World) give geometry from one image, if they hold.

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